Computers and Electronics in Agriculture 156 (2019) 406–425

Contents lists available at ScienceDirect

Computers and Electronics in Agriculture

journal homepage: www.elsevier.com/locate/compag

Original papers

Agricultural technology assessment for smallholder farms: An analysis using a farm simulation model (FARMSIM)

Jean-Claude Bizimana[a,*], James W. Richardson[b]

[a] Department of Agricultural Economics, Texas A&M University, 600 John Kimbrough Blvd/Rm 362 AGLS Blg, College Station, TX 77843-2124, United States
[b] Department of Agricultural Economics, Texas A&M University, 600 John Kimbrough Blvd/Rm 351 AGLS Blg, College Station, TX 77843-2124, United States

ARTICLE INFO

Keywords:
Simulation
Small-scale irrigation
Technology
Risk
Nutrition
Ethiopia

JEL classification:
O1
O2
O3

ABSTRACT

Previous models used to evaluate the adoption of agricultural technologies have focused mainly on assessing the ex-post impact of technology without necessarily quantifying the profit and risk associated with the adoption of technology. This study introduces a farm simulation model (FARMSIM) that evaluates the potential economic and nutritional impacts of new agricultural technologies before and after they are adopted (ex-ante and ex-post). FARMSIM is a Monte Carlo simulation model that simultaneously evaluates a baseline and an alternative farming technology for a representative farm. The farming technologies under study comprise water lifting technologies (pulley and tank, rope and washer pump, gasoline/diesel motor pump and a solar pump) and fertilizers. The study finds that the use of fertilizers combined with optimal irrigation to grow vegetables and fodder using a motor pump had the highest net present value and profit compared to other scenarios. However, the findings also indicate the scenario under the pulley system, a low cost technology, to be the most feasible with the highest benefit cost ratio. Solar pump system had a lower benefit cost ratio compared to the pulley and motor pump scenarios due to high initial investment cost. Based on simulation results, all four water lifting technologies are profitable and suitable for use in Robit kebele. Moreover, due to improved technologies in alternative scenarios, the nutrition simulation results show in general an increase in food quantities available to the family for calories, proteins, iron, fat, calcium and vitamin A. The daily minimum requirements per adult equivalent were met for all nutrients under study except for fat and calcium. Supplemental nutrition through production and purchase, to fill the gap in fat and calcium is recommended for households in Robit kebele.

1. Introduction

For most of the world's poorest countries, especially those on the African continent, agriculture continues to be the main source of employment and contributes to a large portion of the GDP. However, in many Sub-Saharan African countries, including Ethiopia, agricultural productivity has remained low with episodes of famines in drought-prone areas (Gebregziabher et al., 2013; Ahmed, 2015). Issues of food security have surged in recent years and attracted attention from several governments especially those in the developing world where the vast majority of the world's undernourished people are located (FAO, 2001a). To understand why people remain poor and hungry, it is important to know the factors affecting agricultural productivity, which include but are not limited to technologies, resources and institutions that regulate the economy (Norton, 2014; Ahmed, 2015). Farmers in Sub-Saharan Africa need not only to acquire improved agricultural technologies for crop and animal production but also be protected against undue risk related to production.

Adoption of new agricultural technologies leads not only, to increasing productivity and food security, but also enhances agricultural development while reducing poverty. However, risk and uncertainty linked to the process of agricultural production and marketing constitute one of the barriers for farmers to adopt new agricultural technologies. Moreover, adopting and using new agricultural technologies has never been an easy task because of many factors that are involved in the adoption process (Doss, 2006; Diagne et al., 2012). Generally, farmers look at some or all of factors and choose to adopt a technology based on their utility and profit maximization behaviors (Qasim, 2012). The assumption is that farmers engage in adoption of new technology only if the benefits or perceived utility of using the new technology outweighs the benefits of the current or old technology.

Several models, such as the binary choice models have been used to measure the adoption of technologies. However, these models, which do not necessarily quantify the profit and risk associated with

* Corresponding author.
E-mail addresses: bizimanatex@tamu.edu (J.-C. Bizimana), jwrichardson@tamu.edu (J.W. Richardson).

https://doi.org/10.1016/j.compag.2018.11.038
Received 27 February 2018; Received in revised form 1 October 2018; Accepted 26 November 2018
0168-1699/ © 2018 Elsevier B.V. All rights reserved.

production, assess the ex-post impact of technology adoption (de Janvry et al., 2011; Diagne et al., 2012). The key result from these models is the average effect of adoption on outcomes (yields, revenues, profit…) for those who have adopted technologies, also called "the average treatment effect". Because of the selection problem, the main challenge for these models is to establish a proper counterfactual group against which to compare adopters especially in the early stage of adoption where we have large numbers of non-adopters. Another concern in impact assessment using this approach stems from the inability to detect statistically significant differences in poverty-related outcome measures when agricultural technologies generate small increases in yields and income. Also, note the difficulty of capturing the spillover effect from adoption, which affects adopters and non-adopters. The main issue with the average treatment is its variation over time because the adopters change how they use the new technology with time as they learn more about it and a number of late adopters join the early adopters group. Other types of approaches such as the computable general equilibrium models (CGE) have been used to measure the impacts of technology and economic growth, at the macro level in Africa (Diao et al., 2010; de Janvry et al., 2011; Dorosh and Thurlow, 2013). However, the CGE models neither estimate impacts nor capture appropriately the risk associated with agricultural production at the farm level.

The methods discussed above (binary choice and CGE models) for evaluating the impact of technology has mainly focused on the ex-post evaluation. This paper demonstrates a methodology that evaluates the potential impacts of new agricultural technologies before and after adoption takes place (ex-ante and ex-post evaluation) and explicitly captures the risk of adoption. A farm level simulation model (FARMSIM) is used to carry out this task. In addition to assessing and quantifying the economic profit, the model evaluates the nutritional outcomes for a farm family that adopts new agricultural technologies based on the increased consumption and sale of production surplus to buy other foods not produced on the farm. All of these output variables are projected for five years and presented in terms of probability distributions estimated by simulating yield and price risk based on historical variability. Effects related to price elasticities to reflect the changes in revenues and costs from a potential increase in crop production are taken into account by the model. The focus of this paper is to describe the FARMSIM model and demonstrate its ability to evaluate the probable economic and nutritional impacts of adopting agricultural technologies for a farm in a developing country setting. A case study from Robit kebele (a kebele is roughly equivalent to village), in the Amhara region of Ethiopia is used as a demonstration farm.

Ethiopia, as many other African countries, has an agriculture-dependent economy (43% of GDP) that include crop and livestock production (USAID, 2011). However, the Ethiopian economy and specifically its agriculture development sector is very vulnerable to external shocks such as climate shocks. Drought is the most prominent climate shock that affect not only food and livestock production but also livelihoods. Historically Ethiopia has known several droughts but the most memorable remains the one associated with the famine of 1983–1985. The most recent severe droughts occurred in 2010–2011 and 2015, putting at risk several millions of people and causing the loss of several thousand animals (FEWS NET, 2015). Given the loss of crops and livestock due to drought, one of the Ethiopian government priorities is to increase irrigation for crop and livestock (USAID, 2011). However, agriculture and livestock production can only have a substantial growth when irrigation is used alongside other agriculture inputs such as fertilizer and improved seed. Because of the limited availability and usage of irrigation methods and improved seed and fertilizer, agriculture yields for smallholder farmers in Ethiopia are below the Sub-Saharan Africa averages. This paper is organized into three additional sections. First, a brief literature on technology adoption and risk in agriculture is offered, followed by a discussion of our method of analysis and empirical results. Lastly, summary conclusions and suggestions for future studies are provided.

2. Literature review

2.1. Technology adoption and agricultural development

Productivity of smallholder farms in developing countries is known to be constrained by several policy and structural issues that have led to slow increases in crop yields and stagnation (Yengoh et al., 2009; Norton, 2014). The absence of technologies, limited access to or the use of inappropriate technologies have been in part blamed for the lack of increases in productivity and persistent food insecurity in developing countries. Although progress has been made since 2000 to reduce the proportion of hungry people in developing countries (level of hunger was cut by 29 percent), its levels are still alarming especially in Sub-Saharan Africa (von Grebmer et al., 2016). Usually it is assumed that with the right technology (seeds, irrigation, fertilizers, technique, tools), agricultural production will increase and food security will be restored in areas with physical and social limitations to production. However, this has not always been the case. National governments, international agencies and several development organizations have attempted to make agriculture more productive and profitable by introducing agricultural technologies but with modest results (Yengoh et al., 2009). Several factors have contributed to low (or lack) of new technology adoption in developing countries including the limited access to technologies.

For instance, conservation technologies have been introduced in places with steep slopes where erosion was a major limiting factor to production, such as in the Ethiopian Highlands (Gebregziabher et al., 2013). These technologies include soil and water conservation (SWC) techniques, extensively recognized as a key strategy to improve agricultural productivity. However, in the rain-fed agro-ecological landscapes of Ethiopia, the low yield (on average about 35 percent of the potential) is not always due to the lack of water but rather a consequence of inefficient management of water, soils, and crops. Given the modest impact and outcome of these conservation measures, emphasis of the interventions should not only focus on the engineering and biophysical performance of conservation measures but also on the socioeconomic and livelihood benefits (Zemadim et al., 2011). Experience also shows that even when technologies are appropriate for the biophysical setting, they are not always adopted because farmers consider multiple factors when making a decision to adopt a particular intervention. Externally driven technical solutions are rarely sustained by farmers unless consideration is given to socioeconomic, cultural and institutional in addition to biophysical and technical factors.

2.2. Accounting for risk in agricultural technology adoption

Agriculture is often characterized as a relatively high risk and uncertain enterprise in the economy due to a number of risky factors related to the agricultural business such as production, marketing and finance (Hardaker et al., 2004; Di Falco and Chavas, 2009; Qasim, 2012). Some of the risk and uncertainty components surrounding the production and marketing processes are related to unpredictable weather (drought, frost, flood, and wind storm), input quality, pest and disease attacks, price fluctuations, new technology failure, and changes in government policies. These factors are some of the main causes of farm production and income fluctuations at the micro and macro-economic levels. Financial risk is more related to the failure to meet liabilities with the cash generated through farm revenues due to a disparity between cash inflows and outflows, the level of debt, and other sources of financial resources. The combined effect of production, marketing, institutional and personal risk, called *business risk,* is the cumulative effect of all the uncertainty surrounding the profitability of the firm (Hardaker et al., 2004).

There is a need to explicitly take into account risk in agriculture

because farmers cannot control all of the factors related to agricultural production and marketing. Due to unpredictable weather and harsh climatic and agro-ecological conditions, crop production failure can result in food insecurity and famines such as those that have happened in Ethiopia over the last 40 years (Di Falco and Chavas, 2009). With recurring drought and a high risk of harvest failure, ex-ante farm production decisions based on type of crops along with water conservation and irrigation technologies are part of risk management strategies and can help cope with drought risk.

In this paper, risk is explicitly taken into account to study the economic and nutrition impacts of adopting small-scale irrigation (SSI) technologies in Robit kebele. Farm modelling using a Monte Carlo simulation model is one of the approaches that can help to analyse risk and quantify the economic feasibility of adopting SSI technologies on a farm (Richardson et al., 2007). In this regard, stochastic variables that contribute the most to risk in agriculture such as price and yields are simulated to model risk, which is incorporated into the main economic and nutrition key output variables to make risk-based decisions.

3. Data and study area

Robit kebele is one of four intervention sites selected to study small-scale irrigation in Ethiopia, funded under the Feed the Future Innovation Laboratory for Small-scale Irrigation (ILSSI) project and implemented by the International Water Management Institute (IWMI), International Food Policy Research Institute (IFPRI) and the International Livestock Research Institute (ILRI). The case study of Robit used both primary and secondary data for FARMSIM input information. The primary data consisted of a household and community surveys conducted respectively in 2014 and 2015 by the LIVES project[1] (Livestock and Irrigation Value chains for Ethiopian Smallholders) (Gebremedhin et al., 2015) and IFPRI (IFPRI, 2015).[2] The primary data were supplemented by secondary data that included expert opinion, research articles, and reports from government and non-government agencies. The information from survey and other sources relates to crops, livestock, assets, liabilities and costs for a representative farm. The baseline input data for a farm in Robit was developed from a sample of 88 households while the input for the alternative scenarios was drawn from field studies led by IWMI and ILRI in 2015 and 2016.[3]

Robit kebele is located in Bahir Dar Zuria district (woreda), West Gojam zone in Amhara region of Ethiopia approximately 20 km from Bahir Dar town (Fig. 1). The Robit village area has an average elevation of 1848 masl (meters above sea level). According to the 2007 Ethiopia Census results, around 8900 people were living in the kebele (Federal Democratic Republic of Ethiopia, 2008).

A mixed crop and livestock production is the predominant farming system in the area where the main crops grown include maize, finger millet, teff, rice, and chickpeas. Crops are grown using both rain and irrigation water. Two major cropping seasons are identified in Ethiopia: *Kiremt* and *Bega*. *Kiremt* is the main rainy season (June-September) during which major field crops (mainly grains) are grown and harvested in *Meher* season. Irrigated crops such as tomatoes, grass peas, chickpeas, cabbage and onions are grown during the *Bega* season (dry from October to January). The main source of irrigation water is from shallow wells. Most of the households keep cattle, small ruminants, poultry and bees (apiculture). Cattle are basically raised to meet draught power requirements while milk, meat, manure, dung cake, breeding replacement stock are income sources. The majority of the milk produced is retained for home consumption. However, some milk is processed into butter for sale and family consumption. In this study a representative farm from Robit kebele is simulated in the model to identify the impact of new agricultural farming technologies on farm profit and nutrition (see characteristics of a Robit farm in Appendix A).

4. Methods

4.1. FARMSIM model description

The farm simulation model "FARMSIM" is a Monte Carlo simulation model that simultaneously evaluates a baseline and alternative technologies for a farm. The model is programmed in Microsoft ® Excel and utilizes the Simetar© add-in to estimate parameters for price and yield distributions, simulate random variables, estimate probability distributions for key output variables (KOVs) and rank technologies (Richardson et al., 2007).[4] FARMSIM is programmed to recursively simulate a five-year planning horizon for a diversified crop and livestock farm and repeats the five-year planning horizon for 500 iterations.[5] After simulation, the resulting 500 values for each of the key output variables (KOVs) define the empirical probability distributions used to compare the base and alternative farming technologies or interventions. By comparing the probability distributions for the base and alternative technologies, decision makers can quantitatively analyze the probable consequences of introducing alternative technologies.

FARMSIM is programmed to simulate annual production and utilization of up to 15 crops as well as cattle, dairy, sheep, goats, chickens, and swine. The farm family is modeled as the first claimant for crop and livestock production with deficit food production met through food purchases using net cash income from selling surplus crops and livestock production. Standard accounting procedures are used to calculate: receipts, expenses, net cash income, annual cash flows, and net worth. The KOVs for the model can include all endogenous variables in the model but most attention is focused on annual net cash income, annual ending cash reserves, net present value, benefit-cost ratio (BCR), internal rate of return (IRR), and annual family nutrient consumption of protein, calories, fat, calcium, iron, and vitamin A.

4.2. Monte Carlo simulation for technology and risk analysis

Simulation has become an important tool for risk analysis and a decision aid tool to manage risk for farmers and policy makers. To determine the level of risk and its mitigation, we simulate stochastic variables that contribute the majority of the risk: prices and yields. Since these input variables are stochastically dependent, this requires estimating and simulating the joint distribution, which provides a robust stochastic component for the simulation model used to generate the probability distributions for the key output variables (KOVs).

A multivariate empirical (MVE) probability distribution developed by Richardson et al. (2000) to simulate stochastic variables in farm-level models used for policy and strategic planning analyses is used in FARMSIM. The procedure consists of a semi-parametric Monte Carlo simulation technique, which incorporates intra- and intertemporal correlation and allows the researcher to control the heteroscedasticity

[1] Find more information on the survey at: https://lives-ethiopia.org/2014/06/06/baseline-surveys/.

[2] International Food Policy Research Institute (IFPRI), 2015, "ILSSI/IFPRI study on irrigation, gender, and nutrition", https://doi.org/10.7910/DVN/DH1O3J, Harvard Dataverse, V1.

[3] For more information on field studies, see the ILSSI mid-term report at http://ilssi.tamu.edu/media/1389/final-ilssi-mtr-6-dec-16–3.pdf.

[4] FARMSIM is a micro-computer, Excel/Simetar driven, enhanced version of FLIPSIM designed to simulate smallholder farms in developing countries. FLIPSIM has been used extensively for policy analysis and technology assessment for farms in the United States (Richardson and Nixon, 1985).

[5] Extensive testing with the Latin Hypercube sampling procedure in Simetar has shown that a sample size of 500 iterations is more than adequate to estimate a probability distribution for KOVs in a business model with more than 100 random variables; see Richardson et al. (2007) and simulation course textbook (Chap. 4p.6) at https://www.afpc.tamu.edu/courses/643/2017/simulation/Textbook-Simulation-Applied-Risk-Manual.pdf.

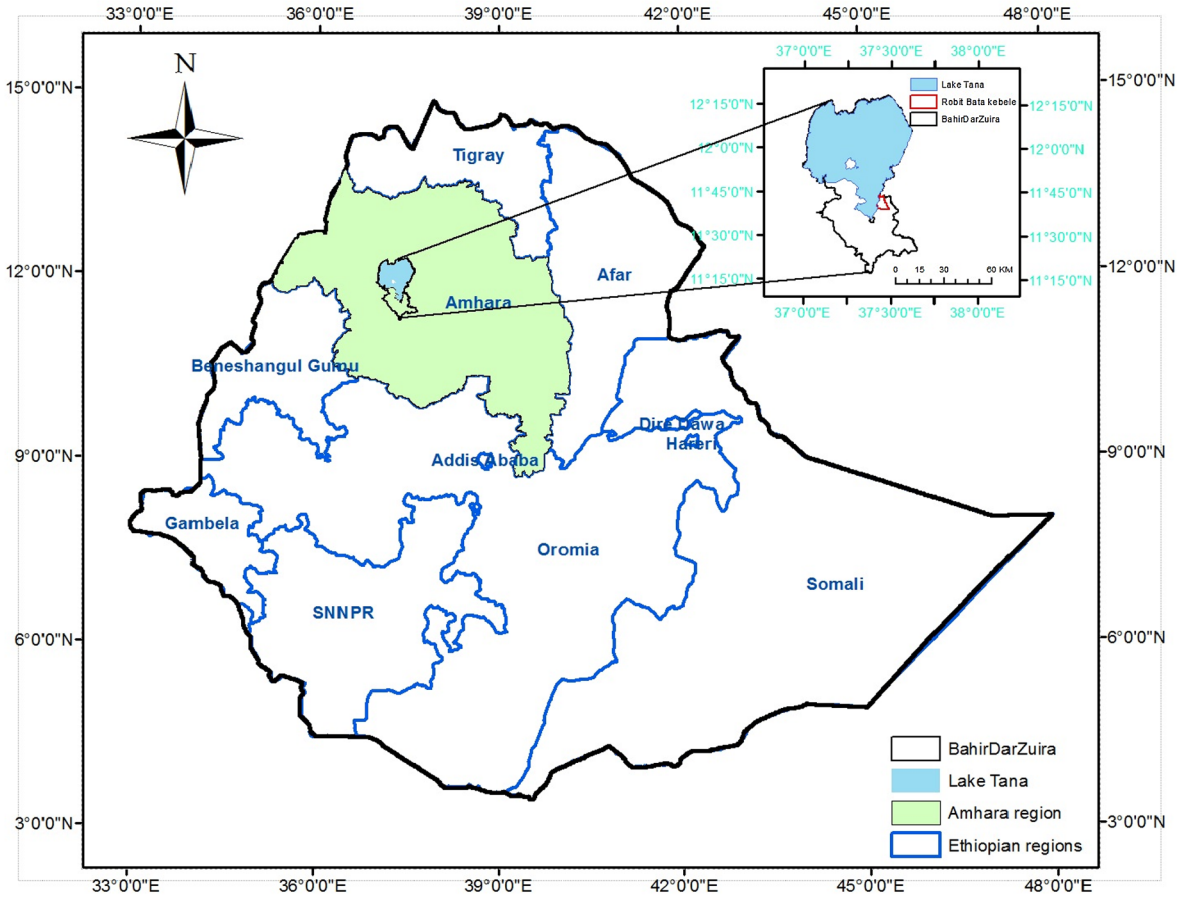


Fig. 1. Location of Robit kebele in Bahir Dar Zuria woreda, Amhara region.

of the random variables over time. (See Eqs. (B.1) and (B.2) in Appendix B). Stochastic annual output prices and yields for crops and livestock are simulated using multivariate empirical (MVE) probability distributions estimated from historical data.[6]

The baseline and alternative technology scenarios are simulated by FARMSIM using the same equations so the only difference in the economic and family nutrition outcomes are due to the technology differences. Random crop yields are simulated using the same stochastic uniform standard deviates to ensure that the weather risk for a crop under the base and alternative technology scenarios are identical. The same stochastic prices for crops are used for both scenarios, unless the alternative scenario calls for a different marketing program, which shifts the price distribution to a different level. Since the baseline and alternative scenarios use identical equations, the decision maker can be assured that the differences in the KOVs are due to the differences in the two farming scenarios and their assumed yield and marketing distributions.

4.3. FARMSIM economic and nutrition equations

The FARMSIM model consists of accounting and nutrition equations that describe annual production, marketing, financial management, household consumption and nutrition. Four major components of the model are presented: crops, livestock, financial and nutrition (see FARMSIM model equations in Appendix B and flowchart in Appendix C).

4.3.1. Crop Model.

The farm family can grow/consume up to 15 different crops/food stuffs. The crops in the model are dependent upon the local crops produced and consumed in the local area. The order of the crops is specified by the analyst so the model is applicable across countries and farming systems. Crop production equals the product of stochastic yield and hectares planted (B.3). Family consumption is calculated as the maximum of average annual kilograms consumed or a fraction of total production. If surplus production exists it can be paid to employees and landowners, where applicable. After satisfying family food needs, employee compensation and rent requirements, the remaining crop production is sold (B.7).

Cash receipts for crops equals quantity sold times the stochastic market price (B.8). Market prices are simulated using either GRKS probability distributions elicited through expert opinion in the absence of historical data or a multivariate empirical (MVE) probability distribution where historical prices are available. As an alternative input, the GRKS distribution (developed by Gray-Richardson-Klose-Schuman) uses three parameters (minimum, a mid-point, and a maximum) and then assigns a piecewise normal distribution such that 50 percent of the density is below the mid-point and 50 percent is above the mid-point.

Variable cash costs of production for crops are calculated as the product of hectares planted and the sum of per hectare costs for seed, fertilizer, chemicals, weeding, irrigation, land preparation, harvesting, and other cash costs (B.9). Fixed costs are inflated by assumed annual inflation rates. Interest costs for land and machinery loans are added to fixed costs for convenience and interest for cash flow deficit loans the

[6] In this study, stochastic annual crop yields were simulated from an MVE probability distributions estimated using 32 years of crop yields generated by APEX (Agricultural Policy / Environmental eXtender) (Williams et al., 1998). A GRKS probability distribution defined below is used where no historical data is available. Selected SSI technologies are simulated by APEX using the same historical weather data and plant growth parameters consistent with the technologies under study.

year before (B.11). Total cash costs for variable crop and livestock costs are charged on interest rate if an operating loan is in effect (B.12). There are no econometric equations in the crop model because all simulated variables are calculated as identities and standard accounting practices.

4.3.2. Livestock model

The livestock component of FARMSIM simulates annual production and herd dynamics for cattle, oxen, chickens, sheep, goats, and swine. Since the cattle sector is the most complex it is described here in detail (B.13)–(B.27). The other livestock sectors are modeled similarly. The number of cows on January 1 less cows consumed, cows die, and cows culled, plus raised replacements, and purchased cows equals the number of cows on December 31. Cows consumed and cows that die are calculated based on relevant fractions for consumption and death and the number of cows on January 1. Cows culled and purchased are endogenous variables calculated annually to maintain the cow herd at the farmer's pre-determined number of cows for each year.

Risk is a significant problem for livestock production. Risk is incorporated into the livestock sector by annually drawing stochastic values for birth rates, death rates, and prices. Death rates are simulated for each age cohort of the livestock specie: yearling calves, two-year-old heifers, three-year-old heifers, mature cows, and bulls (B.14). Stochastic prices are simulated for each age cohort to help calculate receipts from selling (culling) animals at different ages. For the cow herd the model also simulates stochastic milk produced for each mature cow that had a calf.

The number of calves born each year is the product of the number of mature cows on January 1 and the stochastic annual calving rate (B.13) and (B.17). Half the calves are assumed to be heifers and half are bulls. The fraction of calves consumed by the family, die or are sold, decrease the number born to arrive at the number of yearling heifers and bulls on December 31. Different death rate, consumption, and sold fractions are used for bull and heifer calves. The model can either sell bull calves or retain them to raise oxen on the farm.

The number of 12–24 month-old heifers on December 31 equals the number of 12-month old heifers January 1 minus the number consumed, culled, and died (B.19). The same process is used to dynamically simulate the 25–36 months old replacement heifers. It is assumed replacement heifers are bred to calve at 40–48 months, and they enter the cow herd once they have a calf.

The dynamics of the oxen herd is similar to the cow herd. The number of oxen on January 1 is reduced for the fractions consumed, die or are culled (sold). The number of oxen raised or purchased to maintain the desired herd size is added to the net January 1 herd size to calculate the inventory of oxen on December 31.

Milk production is calculated by multiplying stochastic milk per cow and the number of lactating cows (B.28). The annual number of lactating cows equals the number of cows that calved each year. Milk production per cow is a stochastic variable simulated using an expert opinion parameterized GRKS distribution that is augmented by fractional adjustments for forage production. Low (high) forage production reduces (increases) the stochastic milk per cow value in the current year. Low forage production not only affects milk production in the current year but also reduces the stochastic fertility rate. Thus, a drought causes decreased milk production for two years as fewer lactating cows are available in the second year.

Milk consumption by the family is a fraction of total milk production. A fraction of the milk can be made into butter, which can be consumed by the family or sold. Milk not consumed or made into butter is sold. Receipts for cattle are calculated using the stochastic prices for the respective age categories and the number of head sold plus receipts for milk, butter, hides, and manure (B.29). It is assumed only male calves are sold in (B.29). The receipts for hides come from multiplying a price for hides by the number of cattle and oxen that die or are consumed. Manure receipts are calculated as the product of manure prices and the sum of manure produced by cows, calves, replacements, and oxen, if the manure is not used on the farm.

Cash costs for the cattle herd are calculated by multiplying the annual cash costs for cows, calves, two year olds, three year olds, and oxen by the number of head on December 31 (B.10). The annual cash costs are inflated each year by assumed rates of inflation. As indicated above, the livestock model has similar detail for simulating the herd dynamics, receipts, and costs for sheep, goats, and swine. For chickens the production, consumption, and sale of eggs is added to the flock dynamics, which are similar to the other livestock species.

4.3.3. Financial model

The financial model consists of three pro forma financial tables: income statement, cash flow, and balance sheet (B.32)–(B.40). The income statement shows the source of annual receipts from crops, cattle, chickens, sheep, goats, and swine (B.32). Annual cash expenses from crops and livestock plus fixed costs and interest costs are summarized in the income statement (B.33). Annual net cash income equals total cash receipts minus total cash expenses (B.34).

The cash flow statement calculates cash inflows, cash outflows, and ending cash reserves. Cash inflows equal the sum of beginning cash, net cash income, off farm income, and interest earned for cash reserves (B.35)–(B.37). Cash outflows (B.39) equal the sum of cash purchases of food, school expenses, family living expenses, income taxes, principal payments, and repayment of cash flow deficits in the previous year. Ending cash reserves on December 31 (B.40) equals total cash inflows minus total cash outflows and is a KOV for assessing the effects of technology adoption.

The balance sheet summarizes assets and liabilities for the farm (B.41)–(B.46). Annual assets equal the value of land, livestock, machinery, and positive cash reserves. Annual liabilities equal remaining debt for initial loans and new debts, and negative cash reserves. Net worth equals assets minus liabilities (B.46).

Net present value for the farm family equals the present value of annual family withdrawals plus the value of crops and livestock consumed, plus the present value of ending net worth, minus the beginning net worth (B.49). A 10 percent discount rate is used to calculate net present value. Two NPV-related metrics commonly used to assess the economic feasibility and profitability of project or enterprise are as well included in FARMSIM: internal rate of return (IRR) and benefit-cost ratio (BCR). The IRR is defined as the discount rate that causes the NPV to equal zero. BCR is the ratio of the present value of the benefits to the present value of the costs. If the ratio is greater than one, the business or project is profitable. This is an equivalent condition to the NPV criteria where if the discounted present value of the benefits exceeds the discounted present value of the costs then the project is worthwhile.

4.3.4. Nutrition model

The total kilograms of each raised crop consumed by the family plus the kilograms of purchased foodstuffs are multiplied by their respective nutrient scores to calculate total calories, protein, fat, calcium, iron and vitamin A from the food stocks (e.g. see proteins in (B.50)). Similar calculations are made to simulate nutrients derived from consuming cattle, oxen, milk, butter, chickens, eggs, mutton, lamb, nannies, kids, and pig meat. Total nutrients consumed by the family from all sources, including donated food, are summed across plant and animal food stocks and compared with minimum daily recommended amounts for adults based on FAO requirements standards (FAO, 2001a, 2001b, 2010, 2011).

4.3.5. Summary probabilities

The risks associated with the base and alternative scenarios are compared using several different metrics. The probability of a positive net cash farm income (B.51) is calculated by counting the number of iterations that net cash farm income is positive and dividing by 500 (the number of iterations simulated). The probability of a positive net cash

farm income is calculated for each year of the planning horizon. Similar calculations are included to calculate the probability that annual ending cash is positive (B.52), the probability that NPV is positive, which is the probability of economic success (B.53), and the probability that the farm increased real net worth (B.54). The probability that the family daily consumption of protein exceeds the average daily minimum established by the Food and Agriculture Organization and World Health Organization is calculated in (B.55). The calculation is repeated as well for the other nutrients.

5. Technology scenario analysis

5.1. Crop production scenarios

Data input into FARMSIM is entered in parallel for the baseline and alternative scenarios. For each input variable the user must provide information for the current (baseline scenario) and the alternative farming system (alternative scenario) (Appendix C). The model is designed so the user can enter complete data sets for the baseline and up to 21 alternative scenarios. Due to the importance of irrigation in crop and livestock production and recurrent drought episodes observed in Ethiopia, small scale irrigation technologies are discussed using a case study of Robit kebele in the Amhara region of Ethiopia. The scenario analysis consists of the evaluation of water lifting technologies and fertilizer applications. Given that most of the water used for irrigation in Amhara is groundwater from wells, four water lifting technologies ranging from pulley and tank, to rope and washer pump, to motor and solar pumps are evaluated for their capacity and affordability (Appendices D.1 and D.2).

Three major cereal crops (maize, teff and millet) consistent with the cropping system in Robit and grown during the wet season are considered in this study. In addition to cereal crops, chickpeas, potato, cabbage, tomato, fodder and napier grass are analyzed in the model (Appendix E.1). Assuming no need for supplemental irrigation for the cereal crops, chickpeas and potato, the main difference in yield between the baseline and alternative scenarios in terms of technology input, for these crops, would mainly come from fertilizer application.

Literature and a recent household survey carried out in Robit by the LIVES project indicate that a relatively adequate amount of fertilizer (DAP and Urea), close to the recommended rates, is used by farming households (baseline scenario) in Robit for maize and millet (Dile and Srinivasan, 2014). Hence, no additional fertilizers were considered for maize and millet in the alternative scenarios. However, due to low fertilizer rates application for teff in the baseline scenario, increased levels of fertilizers were used in all alternative scenarios. Notice that the increase in yield from the baseline to the alternative scenarios for millet and maize are mainly due to the deep tillage applied in Robit kebele for all alternative scenarios to break the hardpan soil[7]. With regard to potato, increased fertilizer rates were applied in all alternative scenarios compared to the baseline but rates were still considered lower than the minimum recommended levels (USAID, 2013). The same levels of fertilizers (current), but limited as well, were considered in baseline and alternative scenarios for chickpeas since it did not show any stress for phosphorus and has the capability of fixing nitrogen (see details on fertilizer rates in Appendix E 2). The survey information shows as well that most of the households used stored seeds from the previous harvest for the following planting season (except for potatoes in Alt. scenarios) and that the use of chemicals was limited. It was also noted that the level of farm labor hiring for agricultural production in the baseline scenario was low since family members performed some of the agricultural tasks. However, in the alternative scenarios more hired labor was considered, especially for irrigation activities.

The irrigated crops are grown during the dry season and consist mainly of tomato and cabbage in the vegetable category and fodder (oats & vetch) and napier grass in the animal feed category. Note however that a portion of fodder and napier grass production was simulated as a market commodity (cash crop) while the remainder was used to feed animals. While the required fertilizer rates for tomato (Urea: 200 kg/ha and DAP: 100 kg/ha) were applied in the alternative scenarios (USAID, 2013; ILSSI mid-term report, 2016[8]), household data from the LIVES project survey showed only limited application of the current fertilizer rates (Urea: 56 kg/ha; no DAP) in the baseline scenario. In the case of cabbage, given that, few households reported applying fertilizers (baseline scenario), an extremely low amount of fertilizer compared to the recommended rates (for more information on rates, see Gebremeskel, 2016) was considered for the base and alternative scenarios. Therefore, the cabbage yield difference was mainly due to the quantities of applied irrigation water and subsequent water stress levels, which were at 50 percent for the baseline and zero percent for the alternative scenarios. For animal feed (fodder and napier grass), additional amounts of fertilizer to the current levels were applied to fodder (oat & vetch) as well as purchase of seeds in alternative scenarios. Minimal levels of fertilizer and irrigation (when needed) were applied for napier grass in both the baseline and alternative scenarios as no information on the current application was available from the household surveys in Robit. Details on fertilizer rates are in Appendix E.2 while those on crop yields, land allocation and input costs are in Appendix E.3. Five scenarios under study are summarized below:

- Baseline: current fertilizer + current tillage + no or minimal irrigation
- Alt.1 (Pulley): optimal irrigation of vegetables, fodder and napier with pulley & tank + fertilizer + deep tillage
- Alt.2 (Rope & Washer pump): optimal irrigation of vegetables, fodder and napier with rope & washer pump + fertilizer + deep tillage
- Alt.3 (Motor Pump): optimal irrigation of vegetables, fodder and napier with motor pump + fertilizer + deep tillage
- Alt.4 (Solar Pump): optimal irrigation of vegetables, fodder and napier with solar pump + fertilizer + deep tillage.

5.2. Livestock production technologies

Improving animal feed resources can have a positive impact on both household income and nutrition through the production, consumption, and sale of live animals and animal products. In this study, small-scale irrigation (SSI) technologies along with fertilizer application were used to grow and improve yields of fodder and napier grass with a purpose of feeding animals and generating income. Supplementing animal feeding with fodder and napier grass is expected to increase milk production and animal live weight which in turn will improve the family nutrition through milk, butter, and meat consumption and generates income through sales.

Livestock production technologies were aligned with crop production and water lifting technologies (Appendix F). In the baseline scenario fodder crops (oats & vetch) and napier grass are grown on limited land with minimal irrigation and fertilizer applications. However, in the alternative scenarios, more land is allocated to fodder and napier especially during the dry season due to irrigation. Additional land area covered by irrigation for fodder and napier grass varies according to the water lifting technology pumping capacity. Fertilizers are also applied

[7] Issue of hardpan soil in Robit is discussed in this report by the Bahir Dar University (BDU), a partner in the implementation of the ILSSI project: https://ilssi.wikispaces.com/file/view/Mid+Term+Report+Feb+2015.pdf

[8] Report is available at: http://ilssi.tamu.edu/media/1389/final-ilssi-mtr-6-dec-16–3.pdf.

where possible in the alternative scenarios compared to the baseline. A portion of the total production of fodder and napier grass is fed to cows and bulls to increase the production of milk and meat while the remainder is sold for income generation.

Preliminary results on the meat and milk production from a single cut of fodder (oats & vetch) and napier grass were provided by researchers at the International Livestock Research Institute (ILRI) (ILSSI mid-term report, 2016 footnote 7). Assuming all forage is used for production and none for maintenance purposes and considering local cattle breeds feeding with fodder (oats & vetch) and napier grass, there is on average a live weight gain of around 52.4 kg and an improved milk yield of 312 l per year per cow. We assumed also an adoption rate of 60 percent for the livestock technologies, doubling the 30 percent rate of adoption indicated by the ILRI-LIVES household survey. The number of cattle is held constant for the 5 year planning horizon. Following are the baseline and alternative livestock scenarios:

- Baseline: No or minimal irrigation + current animal feeding (no supplemental feed)
- Alt. Scenario 1: Irrigation of fodder & napier w/pulley + supplemental fodder
- Alt. Scenario 2: Irrigation of fodder & napier w/ R &W pump + supplemental fodder
- Alt. Scenario 3: Irrigation of fodder & napier w/motor pump + supplemental fodder
- Alt. Scenario 4: Irrigation of fodder & napier w/solar pump + supplemental fodder.

5.3. Technology scenarios ranking

In addition to evaluating the profitability of SSI technologies, a utility-based approach is used to rank the different alternative scenarios. A range of methods can be used to rank risky scenarios that include means, standard deviation, and coefficient of variation. While some of these approaches include risk, they are not robust enough to always unambiguously, rank scenarios without taking into account the risk preference of the decision maker. Therefore, incorporating utility-based ranking methods to compare alternative farming scenarios is a better approach to help the decision maker select among the scenarios.

Four utility-based ranking functions are included in Simetar and can be used by the FARMSIM model to rank alternative farming scenarios. They comprise the Meyer's Stochastic Dominance with Respect to a Function (SDRF), Certainty Equivalent (CE), Stochastic Efficiency with Respect to a Function (SERF), and Risk Premiums (RP). In this study, SERF was used to rank the risky alternatives given its many advantages over the other methods. Hardaker et al. (2004) merged the use of CE and Meyer's range of risk aversion coefficients to create the stochastic efficiency with respect to a function (SERF) method for ranking risky alternatives. SERF assumes a utility function with a risk aversion range of U (r_1 (z), r_2 (z)) and evaluates the CEs over a range of risk aversion coefficients (RAC) between a LRAC (lower RAC) and an URAC (upper RAC). The range can go from an LRAC equal to zero (risk neutral) to an URAC greater than zero for a risk averse person. In ranking the risky alternatives, the SERF approach compares the CE of all risky alternative scenarios for all RACs over the range and chooses the scenario with the highest CE at each RAC value as the most preferred, and summarizes the CE results in a chart.

5.4. Micro and macro level assumptions

First, to show the full potential of technology adoption, we assumed that the simulated alternative farming technologies are fully adopted. The concern for farmers to acquire irrigation tools such as pumps due to the high capital cost was partially addressed by making available loans to purchase the irrigation pumps through microfinance institutions with the help of ILSSI and IWMI[9]. As for livestock production technologies related to feeding animals with fodder and napier grass supplement, we assumed a 60 percent adoption rate which is two times the original 30 percent adoption rate reported in household data survey. No scientific basis guided this choice apart from the assumption of a scenario where the current adoption rate is doubled.

Second, since the farmer's profit mainly depends on the amount of crop and livestock (including livestock products) sold at the markets, accessibility to markets by the farmers is of paramount importance. Access to markets by farmers depend in part on the existence of road and market infrastructure in the Bahir Dar Zuria district where survey indicates an average 1.4 km (0.9 miles) distance to market. With regard to market existence and operations for instance, a market research report shows that milk was sold at Bahir Dar surrounding markets with a higher demand than the supply (Wondatir et al., 2015).[10] The report indicates as well that supplemental feeds were purchased by farmers at both local and Bahir Dar markets. To maintain quality, vegetables are often sold at the farm gate, roadside and local markets in Bahir Dar. A 2017 household survey in Robit conducted by IWMI on behalf of the ILSSI project indicates as well that about 70% of the vegetables produced (mainly garlic and tomatoes) were sold by farmers at the market (see excel data file in supplemental materials). However, in many developing countries markets fail due to the lack of communication and transport infrastructure, and appropriate regulations and finance (Barrett, 2005; Bizimana et al., 2015). We recognize that the issue of market access and participation is complex in developing countries where several types of transaction costs come into play (Key et al., 2000); these forces are included in the analyses to the extent possible by using prices derived from farmer surveys rather than published average prices for the country.

6. Simulation results and discussion

A baseline and four alternative scenarios are simulated and forecasted for a five-year period using FARMSIM to evaluate agricultural and livestock technologies in Robit kebele. Due to limited space, only the year five simulation results are presented in this section using graphical and analytical tools to rank the risky alternatives scenarios in comparison to the baseline. Cumulative distribution function (CDF), probability density function (PDF) and StopLight charts are presented to compare the scenarios in terms of their financial and nutrition outcomes.

6.1. Net present value (NPV)

NPV is a financial indicator that assesses the feasibility and profitability of an investment or project over a certain period of time. Overall, the NPV simulation results as illustrated by the cumulative distribution function (CDF) indicate that it is worth investing in irrigation and fertilizer application (Appendix G.1). The application of fertilizers on grain and vegetable crops together with the irrigation of vegetables and fodder using a pulley/tank, rope-and-washer, motor, or solar pumps (Alts. 1, 2, 3, and 4, respectively) showed outstanding performance relative to the baseline. Their CDF values lie to the right of the baseline scenario for all 500 draws of the simulation model. The motor pump scenario (Alt. 3) has the highest NPV values at all probability levels compared to the pulley/tank, rope-and-washer and solar pump scenarios.

6.2. Net cash farm income (NCFI)

Annual NCFI measures the amount of profit generated by the farm for the baseline and alternative scenarios. The simulation CDFs of NCFI

[9] See ILSSI mid-term report (2014–2016), p.22 at http://ilssi.tamu.edu/media/1389/final-ilssi-mtr-6-dec-16–3.pdf.

[10] See the ILSSI report at: http://ilssi.tamu.edu/media/1372/ilssi-robit-bata-feast-report-updated-final-2.pdf.

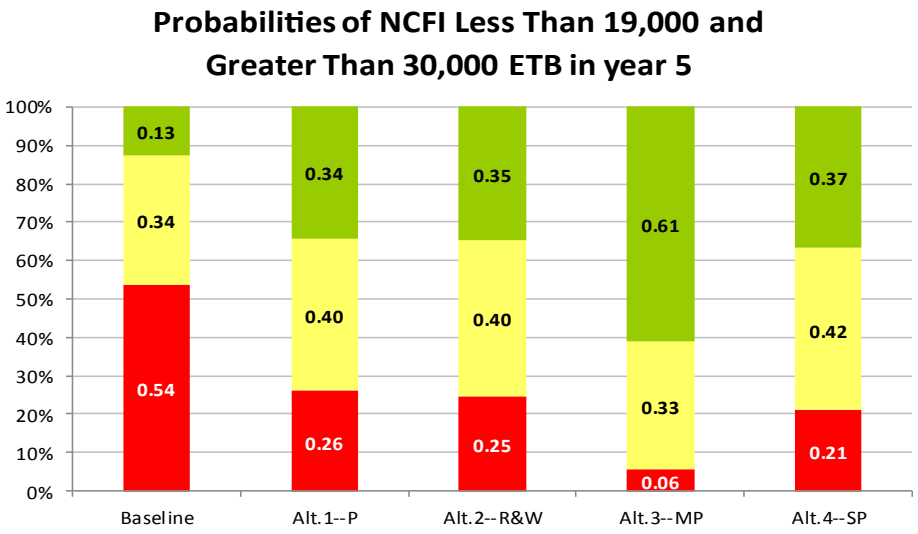


Fig. 2. StopLight chart for per family NCFI in Robit kebele. Legend: Baseline: No irrigation; Alt.1–P: Pulley and Tank; Alt.2–R&W: Rope & Washer pump; Alt.3–MP: Motor pump; and Alt.4–SP: Solar pump.

indicate that the motor pump scenario (Alt. 3) generated higher NCFI than any other scenario at each probability level, followed by the solar pump, rope-and-washer pump and the pulley/tank scenarios (Alts. 4, 2 and 1, respectively) (Appendix G.2).

The StopLight chart in Fig. 2 displays the probabilities of NCFI being less than 19,000 ETB (Ethiopian Birr) (red), greater than 30,000 ETB (green), and between the two target values (yellow) in year five. The target values are average NCFI for the lowest-performing scenario (baseline scenario) for the lower bound and the highest performing alternative scenario (Alt. 3) for the upper bound. The NCFI simulation results show that, for a representative farm or household in the baseline scenario, there is a 54 percent probability that NCFI will be less than 19,000 ETB and a 13 percent probability that NCFI will exceed 30,000 ETB (Fig. 2). In contrast, for the motor pump scenario (Alt. 3), there is a 61 percent chance that annual NCFI will exceed 30,000 ETB, 33 percent probability that NCFI will fall between 19,000 and 38,000 ETB, and a 6 percent that the NCFI will be less than 19,000 ETB. All the other three alternative scenarios (Alts. 1, 2 and 4) performed better and generated higher profits than the baseline scenario. They recorded on average a 35 percent probability that NCFI will exceed 30,000 ETB, 40 percent chance that the NCFI will fall between 19,000 ETB and 30,000 ETB, and a 24 percent probability that annual NCFI will fall below 19,000 ETB. It is worth mentioning that higher NPV and profit values from the alternative scenarios are associated with the expansion of irrigated land and the increase in production and sale of crops and livestock products at the market.

6.3. Benefit-cost ratio (BCR) and internal rate of return (IRR)

Although an alternative farming technology such as a motor pump may generate the highest income and profit amount, it may not have the highest economic return on investment. For this reason, two commonly used NPV-related metrics, the BCR and IRR, were calculated to evaluate the economic feasibility of the different alternative technologies.

Any alternative scenario in the study that shows a positive average value of the BCR (> 1) and an average value of IRR greater than the discount rate (> 0.10) would be deemed feasible and profitable as compared to the baseline scenario (both metrics are based on the present value of the difference). The average values in Table 1 show that all four alternative scenarios have positive BCR values with the pulley leading other alternative scenarios with a BCR value of 5.5. Notice that the solar pump has the lowest BCR value (1.9) among the alternatives. This outcome is explained in part by the level of investment costs (irrigation tool) which is higher for the solar pump and lower for the pulley in comparison to the irrigation benefits. Although the motor pump has the highest net profit, its BCR value is ranked second alongside the rope and washer pump scenario. A similar outcome is observed for the IRR with the pulley showing a higher return to investment (1.6) compared to other alternative scenarios (Alt. 2 & 3 average IRR of 1.1). The solar pump has the lowest return on investment of 0.4.

6.4. Baseline and alternative scenarios ranking

NPV and NCFI simulation results indicate the predominance of the motor pump scenario compared to other alternative scenarios (pulley, rope and washer pump and the solar pump) and the baseline. To complete the scenario ranking, stochastic efficiency with respect to a function (SERF), is used to compare and rank the different scenarios based on the decision maker (DM) utility for income and risk. In addition to comparing income average values, this approach considers all levels of risk aversion for a typical decision maker: from risk neutral, to extremely risk averse. The scenarios are ranked based on their certainty equivalence (CE) value for profit at alternative levels of risk aversion. The motor pump scenario (Alt 3) is the preferred scenario at all levels of risk aversion (Fig. 3).[11] The next most preferred scenarios are the pulley/tank, rope & washer pump, and solar pump scenarios (Alts. 1, 2 and 4). Notice in Fig. 3 the CEs for profit for all scenarios decrease as the level of risk aversion increases, which indicates the decision maker's willingness to take less profit to shield against high risk.

The SERF option in Simetar produces as well a risk premium (RP) chart. The RP can be interpreted as the extra amount of money (profit) the decision maker would get to move from the baseline (risk-free scenario) to an alternative irrigation technology (risky scenario). The RP chart shows the perceived premium that each risky scenario provides relative to the baseline scenario at different risk aversion levels

[11] The range of risk aversion coefficient (RAC) described above which goes from 0 (risk neutral) to r (r = 4/Beginning Net Worth of the household) is divided into three broad categories of risk aversion (risk neutral, normal risk aversion and moderately risk aversion) to help represent the level of risk for a decision maker. Several authors: Arrow (1965), McCarl (1989), Anderson and Hardaker (1992) have suggested several ways to compute RAC values and their corresponding level of risk aversion but without converging on a single approach.

Table 1
Economic impacts of SSI technologies in Robit kebele.

	Baseline	Alt.1–Pulley	Alt.2–R&W	Alt.3–MP	Alt.4–SP
Averages values in ETB /family:					
Net present value (over 5 years)	129,925	176,956	177,752	209,895	180,138
Avg. net profit (in year 5)	19,216	26,559	26,750	34,195	27,321
% change profit: Alt./Baseline		38%	39%	78%	42%
Avg. Benefit-Cost Ratio: Alt/Baseline		5.5	3.9	3.9	1.9
Avg. Internal Rate of Return: Alt/Baseline		1.6	1.1	1.1	0.4

Baseline: No or minimal irrigation; Alt.1–Pulley: Pulley-tank used in optimally irrigated systems; Alt.2–R&W-P: Rope & Washer pump used in optimally irrigated systems; Alt.3–MP: Motor pump used in optimally irrigated systems; and Alt.4–Solar-P: Solar pump used in optimally irrigated systems.

(Appendix H). A positive RP over the range of risk aversion levels for an alternative scenario means its preference over the baseline while a negative RP would mean the preference of the baseline scenario over the alternative scenario. The motor pump scenario (Alt. 3) has a higher risk premium compared to other alternative and the baseline scenarios. Notice also that all alternative scenarios have a positive RP compared to the baseline.

6.5. Nutrition simulation results

In general, the adoption and proper use of agricultural technologies contribute to an increase in the quantity and variety of crops produced. The implications for family nutrition vary according to the types of crops grown and consumed. Also surplus crops can be sold and resulting revenues used to buy food items needed to complement nutrition requirements. Although food purchase options to improve nutrition are not discussed here, the LIVES baseline survey shows that households in Robit kebele did not purchase large quantities of food or receive any food aid as a supplement to their production and nutrition. Note that the nutrition simulation analysis of a farm in FARMSIM strictly reflects the availability and accessibility by the farm family for six food nutrients mentioned above and listed in Table 2 below.

An assessment of the current food consumption by a representative household in Robit shows on average a cereal-based diet dominated by teff and maize, representing 63 percent of total quantity of food items consumed by a family in a week. The consumption of vegetable and fruits represents about 5 percent of total quantity while pulse (beans and peas) account for 9 percent. The survey shows that animal products were not widely consumed. Based on principles of good nutrition characterized by diet diversity, the survey results indicates a lack of variety of food consumed and hence a low quality nutrition. Diets predominantly based on cereals but poor in other nutrients are typical of food insecure households and contribute to malnutrition (Arimond and Ruel, 2004). Although the summary provides insights into the average household diet in Robit, a conclusion cannot be drawn given a small sample size of households reported. This study considers different groups of food and animal products ranging from maize, teff, millet, cabbage, tomatoes, chickpeas, potatoes to meat that are raised and consumed on farm in addition to limited purchases of chicken, beef, goat, potatoes, barley and wheat.

Nutrition results indicate that the quantities of crops and livestock products available for consumption to families in both the baseline and alternative scenarios met and exceeded the minimum daily requirements per adult equivalent (AE) for calories, proteins, iron and vitamin A but were insufficient for calcium and fat (Table 2). The fraction of crop consumed by the family, reported in the household survey, was kept constant for the baseline and alternative scenarios for all crops except for the irrigated vegetables of which the majority was sold at the market. Therefore, the increase in crop and livestock production due to improved farming technologies in alternative scenarios led to an increase in available nutrients (Table 2).

The calories intake simulation results for a representative household in Robit kebele indicate an average daily intake available of 2770 and 3680 calories, respectively for the baseline and alternative scenarios which is significantly higher than the daily minimum requirement of 1750 calories per adult equivalent (AE) in Ethiopia (Table 2). Large land allocation to grain crops compared to other crops (Appendix A and E.1), a characteristic of subsistence farming, and the use of fertilizer contributed to a higher grain production and mitigated any deficiency in energy and calories for both the baseline and alternative scenarios. Moreover, survey information shows that, on average, 68 percent of all grains produced by households in Robit kebele are consumed at home.

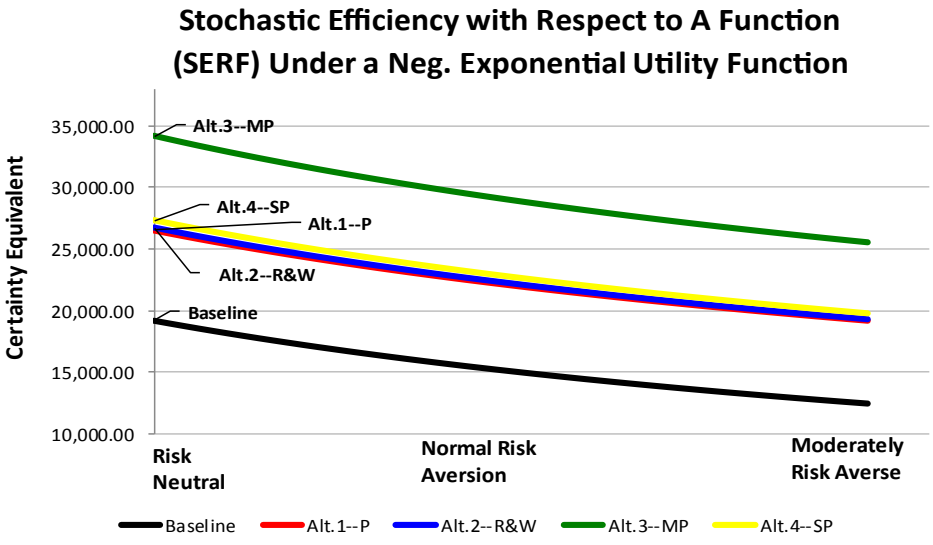


Fig. 3. SERF ranking of alternative farming systems in Robit kebele.

Table 2
Nutritional outcome of SSI technologies in Robit kebele.

	Baseline	Alt.1–Pulley	Alt.2–R&W	Alt.3–MP	Alt.4–SP	
Averages daily nutrients in year 5						**Min. required**
Energy (calories/AE)	2,776	3,682	3,683	3,721	3,686	1,750
Proteins (grams/AE)	67.6	91.0	91.1	92.9	91.2	41
Fat (grams/AE)	**28.7**	**37.5**	**37.5**	**37.9**	**37.5**	39
Calcium (grams/AE)	**0.19**	**0.35**	**0.35**	**0.38**	**0.35**	1
Iron (grams/AE)	0.019	0.027	0.027	0.028	0.027	0.009
Vitamin A (grams/AE)	0.0026	0.0061	0.0061	0.0065	0.0061	0.0006

Note: For nutrition variables: numbers in bold show quantities of nutrients intake < minimum required; AE = Adult Equivalent. The value for minimum requirement in energy (1750 calories/AE) comes from an updated document for an adult in Ethiopia published by FAO (2011).

The simulation results indicate that consumption of maize, millet and teff provided about 82% of total calories in both the baseline alternative scenarios for the household in Robit kebele. Grain or cereal crops represent the basic staple food and a source of calories in developing countries with agriculture-based economies, including Ethiopia.

The protein intake simulation results for both the baseline and alternative scenarios (67 and 93 g/AE respectively) show that on average households' available proteins meet and exceed the daily minimum requirement of 41 g/AE. There is a significant improvement in protein intake available for the alternative scenarios compared to the baseline scenario. However, household surveys showed that the majority of the proteins consumed in Robit kebele were obtained from crops rather than animal products. Simulation results indicate that maize, teff and millet together provided about 75% of total proteins intake available for the household with animal products (meat, milk and eggs) contributing roughly three percent.

Simulation results for fat show a deficit in fat intake for both the baseline and alternative scenarios. Although there is an increase of fat intake available between the baseline and the alternative scenarios, their respective averages, 28 and 37 g, are below the daily minimum fat requirement average of 39 g for an adult. As noticed with other nutrients, the grains (millet, teff and maize) provided about 79% of the total available household fat intake with maize alone contributing about 61 percent of total fat intake.

A similar deficit was observed for calcium intake quantities. The simulation results for calcium show large deficits in calcium intake in both the baseline and alternative scenarios. The average calcium intake per AE is around 0.19 and 0.35 g, respectively, for the baseline and four alternative scenarios, falling short of the daily minimum requirements of 1 g per AE. Note however the significant improvement of calcium intake from the baseline to the alternative scenarios which almost doubled. Unlike the other nutrients discussed above, calcium came mainly from the consumption of milk and teff, contributing about 43% of total available calcium intake, with teff alone contributing about 30% of total intake. A deficit in availability and access to calcium and fat is mainly due to low consumption of animal source foods, which can effectively close the nutrition gap in fat and calcium intake (Shetty, 2010). The large and consistent gap in calcium intake in the current study reflects the existing concern regarding low calcium intake observed in developing countries (vs. developed countries) due to low animal products access and consumption (FAO, 2001b; Agueh et al., 2015).

As for iron, simulation results indicate that households in Robit kebele consume more than the required minimum levels of iron. The average iron intake per AE of all scenarios, estimated at 0.023 g (or 23 mg), was more than two times greater than the daily minimum requirement of 0.009 g (or 9 mg) per AE. There was also a significant improvement between the baseline and the alternative scenarios for iron intake, which averaged 0.19 and 0.27 g, respectively. Maize was the main provider of iron, contributing around 51% of the total available iron intake to the household followed by teff and millet, which contributed 29 percent.

Finally, the simulation results for vitamin A intake indicate adequate to surplus vitamin A intake levels in both the baseline and alternative scenarios. The average levels of vitamin A intake available for the baseline (0.0026 g) and the alternative scenarios (0.0064 g) are 4–10 times higher than the daily minimum requirement for an adult equivalent (0.0006 g). Potato and cabbage were the main sources of Vitamin A, contributing around 88% of the total vitamin intake available to the household.

Simulated levels of nutrition variables (calories, proteins, fat, calcium, iron and vitamin A) available to farm families increased substantially in the alternative scenarios because of production increases due to farming technology (fertilizer and irrigation) and crop diversification. The representative farm grew seven different crops ranging from grains (maize, teff and millet), vegetables (tomatoes and cabbage), root crops (potatoes), pulse (chickpeas), and animal feed (fodder and napier). Although the animal feed crops were not directly used for human consumption they were fed to livestock for milk, butter and meat production used in human consumption. The increase in yields allowed farm families to produce sufficient food for consumption, which provided different nutrients needed for human consumption except for fat and calcium. However, with improvement in livestock production, the deficiencies in fat and calcium can be filled by the consumption of milk, eggs, cheese, butter and meat. Hence, food supplements either through purchase or farming to increase the intake for fat and calcium will be needed to meet the nutritional requirements and the well-being of the families in Robit kebele.

7. Conclusions and recommendations

The purpose of this study was to evaluate the impacts of adopting agricultural technologies (increased fertilizer use and irrigation) on household nutrition and farm profitability in Robit kebele, Amhara region of Ethiopia. A baseline scenario with current fertilizer application rates and no or minimal irrigation was compared to four alternative scenarios where increased fertilizers rates and irrigation were applied to crops. In the alternative scenarios, increased fertilizer rates were applied only to grow teff and potatoes since household surveys showed adequate amounts of fertilizer for millet and maize. Tomato and fodder (oats & vetch) received increased levels of fertilizers as well and were irrigated during the dry season using four alternative water-lifting technologies that include pulley/tank, rope and washer, motor, and solar pumps. Very small amounts of fertilizers were applied to cabbage in the baseline and alternative scenarios but optimal irrigation was considered only in the latter. Napier grass received minimal amount of fertilizers and was irrigated as needed.

The scenario consisting of applying fertilizers in combination with the use of motor-pump to irrigate tomato, cabbage, fodder, and napier grass, generated the highest income and profits for the farm. The next-best performing scenarios used pulley/tank, rope-and-washer or solar pump for irrigation in combination with fertilizer use. It is worth noting that although the solar pump had similar outcome in terms of profit as the pulley/tank and the rope and washer pump, it had a lower benefit

cost ratio due to high investment costs. The baseline scenario was the least preferred of all five scenarios analyzed. The cost-benefit analysis indicated the pulley system to have the highest return due to its low cost of investment. Depending on the irrigation tool affordability and socio-economic status, households may choose between motor pump, rope and washer pump and pulley/tank tools, which are all profitable options but require different levels of investment costs. Although solar technology is the most expensive of all tools, it may be a tool of the future given its environmental cleanness and low maintenance cost as opposed to motor pump which has high maintenance cost and is less environmental friendly.

The use of improved farming systems due to fertilizer and irrigation had a positive but modest impact on livestock production, specifically meat and milk output as well as their consumption at the household level. Available nutrients levels in calories, proteins, iron, fat, calcium and vitamin A improved significantly in the four alternative scenarios compared to the baseline scenario as a result of improved crop yields and variety of irrigated crops. However, simulation results showed consistent deficiencies in fat and calcium. Increased use of fertilizer and irrigation has a potential to increase profit, and improve family nutrition through production and/or purchase and fill the nutrition gaps in fat and calcium in Robit kebele.

Further studies are recommended to elicit how household produce are sold on the market, the level of market demand and supply for food and animal products in and around Bahir Dar and Robit. There is a need as well for additional studies on crops and livestock products with high nutrition value that can be targeted through production and/or purchase to help improve nutrition in Robit.

Acknowledgment

The authors wish to acknowledge the following agencies and individuals who were instrumental in providing data and expert advice. The Innovation Lab for Small Scale Irrigation (ILSSI) team at Texas A& M University: Abeyou Wale Worqlul, Yihun Dile Taddele, Raghavan Srinivasan, Tom Gerik and Neville Clarke; Brian Herbst of the Agricultural Food Policy Center/TAMU; the International Livestock Research Institute (ILRI); the International Food Policy Research Institute (IFPRI); the International Water Management Institute (IWMI). Special thanks to Dr. Azage Tegegne and Dr. Berhanu Gebremedhin from the LIVES project, in Ethiopia for sharing Robit household survey data. The authors would like to, sincerely thank the anonymous reviewer for the helpful and detailed comments.

Funding

This research was made possible and funded by the United States Agency for International Development (USAID), Washington, DC [Grant number: AID-OAA-A-00055].

Appendix A. Farm and village characteristics

A.1. Smallholder farm characteristics in Robit kebele

*Average farm size (ha):			
-cropland	0.91		
-pastureland	0.26		
*Current crops (ha/kebele):			
-Maize	728		
-Millet	708		
-Teff	266		
-Tomato	102		
-Fodder (oats & vetch)	86		
-Onion	43		
-Green Pepper	40		
-Chickpeas	57		
-Potato	23		
*Livestock (# heads/kebele):			
-Cows	2805		
-heifers	1652		
-bulls	827		
-Calves	330		
-Oxen	2974		
-Goats	397		
-Sheep	6196		
-Chicken	21,649		
-Pigs	0		
*Assets (Birr/household):			
-Value tools & Building	7444		
*Liabilities/household	Amount (Birr)	Interest (%)	Terms (year)
-Land loan	0	0	0
-Technology loan	1750	18	1
-Miscellaneous loan	2625	20	1

Appendix B. FARMSIM model equations

The equations in FARMSIM are summarized as follows:

Stochastic annual yields are simulated as:

$$\tilde{Y}ield_{it} = \bar{Y}ield_{it} * (1 + MVEMP(S_y, F(x), CUSD_{yt})) \quad \text{(B.1)}$$

$\bar{Y}ield_{it}$ *is mean yield of crop i in year t*,

MVEMP is the multivariate empirical function,

S_y *is the matrix of fractional deviations from mean yields for all crops,*

$F(x)$*is the cumulative distribution 0 to 1 for the* S_y *matrix, and*

$CUSD_y$ *is a vector of correlated uniform standard deviates for crop yields in year t.*

All equations are repeated for both the base and alternative scenarios. In the case of crop yield each scenario has a different S_y matrix and $\bar{Y}ield_{it}$ vector to match the technology, but use the same independent standard normal deviates to create the $CUSD_y$ vector so both scenarios experience the same weather shocks. The model allows the farm to have up to 15 crops which can be made up of field crops and kitchen garden crops.

Stochastic annual crop prices are simulated as:

$$\tilde{P}riceC_{it} = \bar{P}riceC_{it} * (1 + MVEMP(S_p, F(x), CUSD_{pt})) \quad \text{(B.2)}$$

$\bar{P}riceC_{it}$ *is mean price for crop i in year t,*

MVEMP is the multivariate empirical function,

S_p *is the matrix of fractional deviates from mean prices for all crops,*

$F(x)$ *is the cumulative distribution 0 to 1 for the* S_p *matrix, and*

$CUSD_{pt}$ *is a vector of correlated uniform standard deviates for crop prices in year t.*

The mean price vectors can differ between the base and alternative technology scenarios if one scenario assumes a different marketing option or a different quality of the crop. The multivariate empirical methodology used by FARMSIM is described in detail in Richardson et al. (2000).

Production of each crop i is simulated as:

$$\tilde{P}rod_{it} = Acres_{it} * \tilde{Y}ield_{it} \quad \text{(B.3)}$$

$Acres_{it}$ is the land area devoted to the crop and can be acres, hectares, or any other local name for land area. The model simulates the production identity using stochastic yield so production for each crop is stochastic given weather shocks.

The quantity of the crop which can be sold is the residual after subtracting out the quantity consumed by the family and livestock.

Family consumption and livestock feed requirements are identities that are simulated as:

$$FamilyConsumption_{it} = QF_i * No.\ AdultEquivalent \quad \text{(B.4)}$$

Qf_i is the minimum quantity of crop i consumed per adult equivalent per year and
No. Adult equivalent is the number of adult equivalents in the farm family.

$$LivestockFeed_{it} = \sum_j (QL_{ij} * No.\ Head_{jt}) \quad \text{(B.5)}$$

QL_{ij} is the quantity of crop i fed to livestock type j each year and
No. $Head_{jt}$ is the number of adult equivalent animals of type j where j is the type of livestock: cattle, swine, chickens, sheep, and goats.

Quantity of each crop sold is simulated in two steps:

$$\tilde{P}rodR1_{it} = \tilde{P}rod_{it} - FamilyConsumption_{it} \quad \text{(B.6)}$$

$$\tilde{P}rodSold_{it} = MAX(0, (ProdR1_{it} - LivestockFeed_{it})) \quad \text{(B.7)}$$

Cash receipts for crops are simulated as:

$$\tilde{C}R_{it} = \tilde{P}rodSold_{it} * \tilde{P}riceC_{it} \quad \text{(B.8)}$$

Variable costs for each crop i are simulated as:

$$VC_{it} = \sum_j (Acres_{it} * V_{ijt} * (1 + InflRate_{jt})) \quad \text{(B.9)}$$

VC_{ijt} is total variable costs for each crop i in year t,
V_{ij} is the variable cash cost per acre (land unit) for input j applied to crop i, where j represents seed, fertilizer, herbicides, irrigation fuel, labor, marketing, etc., and
Infl $Rate_{jt}$ is the annual inflation rate in the price per unit of $input_j$ for year t.

Variable costs for each livestock type are simulated as:

$$VL_{jt} = \sum_j (No.\ Head_{jt} * V_{ij} * (1 + InflRate_{jt})) \quad \text{(B.10)}$$

VL_{jt} is the total variable cost for each type of livestock j, and
V_{ij} is the variable cash cost per adult animal of type j.

Fixed cash costs for the farm are simulated by inflating the farm's initial values and summing over the multiple categories of costs.

$$FC_t = \sum_k (FC_{kt-1} * (1 + InflRate_{kt})) \quad \text{(B.11)}$$

Fc_{kt} is fixed cost for k categories which include expenses for: Land leasing, property taxes, insurance, interest for loans, etc.

Total cash costs for the farm family are:

$$TC_t = \left[\sum_i (VC_{it}) + \sum_j (VL_{jt}) + FC_t\right] * (1 + InterestRate) + Interest\ \ Payments\ \ for\ \ Land\ \ Loan + Interest\ \ Payments\ \ for\ \ Machinery\ \ and\ \ Livestock\ \ Loans + Interest\ \ for\ \ Cash\ \ Flow\ \ Deficit\ \ Loan_{t-1} \quad (B.12)$$

The interest rate in Eq. (12) is the rate charged for the operating loan if one is in place. Livestock production is simulated similarly for cattle, swine, sheep, goats, and chickens with exceptions for milk production and the biological progression from infants to adults (calves to cows). Since cattle is the most complex livestock sector, its equations are presented in detail.

The user must specify the number of cattle in four age groups (i.e., cows, 3 year-old heifers, 2 year-old heifers, and 1 year-old heifers). Calving fraction is a stochastic variable simulated using an empirical probability distribution.

$$\widetilde{CP}_t = \bar{CP}_t * (1 + EMP(S, F(x), USD_t)) \quad (B.13)$$

$\bar{CP}_t$ is the average calving fraction such as 0.50 meaning that on average half of the adult cows have a calf.

Death rates for cattle are stochastic and are simulated for each age cohort and for bulls as a multivariate empirical distribution:

$$\widetilde{DR}_{nt} = \bar{DR}_{nt} * (1 + MVEMP(S, F(x), CUSD_t)) \quad (B.14)$$

$\bar{DR}_{nt}$ is a vector of average death rates for cows, heifers 3, 2, and 1 years of age, calves, and bulls.

Milk per cow is a stochastic variable simulated using an empirical probability distribution:

$$\widetilde{M/C}_t = \bar{M/C}_t * (1 + EMP(S, F(x), USD)) \quad (B.15)$$

$\bar{M/C}_t$ is average milk per cow.

Prices for cattle and milk as well as all livestock types are simulated as a multivariate empirical probability distribution for each technology scenario.

$$\widetilde{PriceL}_{jt} = \bar{PriceL}_{jt} * (1 + MVEMP(S, F(x), CUSD_t)) \quad (B.16)$$

$\bar{PriceL}_{jt}$ is the average price vector for all livestock types by age cohort and product such as milk and eggs.

Number of calves born is simulated as:

$$\widetilde{Calves}_t = Cows_t * \widetilde{CP}_t \quad (B.17)$$

Number of calves that die during the year is simulated as:

$$\widetilde{CalvesDie}_t = \widetilde{Calves}_t * \widetilde{DRcalves}_t \quad (B.18)$$

Number of heifer calves enter 1 year-old heifer herd:

$$Heifers1_t = (\widetilde{Calves}_t - \widetilde{CalvesDie}_t) * 0.5 \quad (B.19)$$

Number of 1-year-old heifers enter 2 years-old herd.

$$\widetilde{Heifers2}_t = \widetilde{Heifers1}_{t-1} * (1 - \widetilde{DRHeifers1}_t) * (1 - \widetilde{DRHeifers1}_t) \quad (B.20)$$

Number of 2 years-old heifers enter 3 years-old herd.

$$\widetilde{Heifers3}_t = \widetilde{Heifers2}_{t-1} * (1 - \widetilde{DRHeifers2}_t) * (1 - \widetilde{DRHeifers2}_t) \quad (B.21)$$

Number of cows culled is simulated as:

$$\widetilde{CowsCulled}_t = \widetilde{Cows}_{t-1} * (1 - CowCullFraction) \quad (B.22)$$

Number of adult cows that calve during the year is simulated as:

$$\widetilde{Cows}_t = (\widetilde{Cows}_{t-1} + \widetilde{Heifers3}_{t-1} - \widetilde{CowsCulled}_t) * (1 - \widetilde{DRCows}_t) \quad (B.23)$$

Number of bulls die is simulated as:

$$\widetilde{BullsDie}_t = No.\ Bulls_t * \widetilde{DRBulls}_t \quad (B.24)$$

Number of bulls culled is simulated as:

$$BullsCulled_t = No.\ Bulls_t * BullCullFraction \quad (B.25)$$

Number of bulls to purchase is simulated as the difference between January 1 number of bulls and the number that die and are culled during the year:

$$BullsBuy_t = No.\ Bulls_t - \widetilde{B}ullsDie_t - BullsCulled_t \tag{B.26}$$

$No.\ Bulls_t$ is the number of herd sires at the start of the year.

It is assumed that the number of bulls remains constant over the planning horizon so the number of bulls at year end is:

$$No.\ Bulls_{Dec31,t} = No.\ Bulls_{Jan1,t} - BullsDie_t - BullsCulled_t + BullsBuy_t \tag{B.27}$$

Milk production is dependent on the number of cows that calved and milk per cow or:

$$\widetilde{M}ilk_t = \widetilde{C}ows_t * \widetilde{M}/C_t * \widetilde{C}P_t * ForageIndex \tag{B.28}$$

Cash receipts for cattle are calculated as:

$$\widetilde{C}RCattle_t = (\widetilde{P}Cows_t * \widetilde{C}owsCulled_t) + [\widetilde{P}riceCalves_t * (Calves_t - CalvesDie_t) * 0.5] + [PriceMilk_t * Milk_t * (1 - FractionConsumed_t)] + (BullsCulled_t * PriceBulls_t) \tag{B.29}$$

Variable costs for cattle are calculated as:

$$\widetilde{V}CCattle_t = \widetilde{C}ows_t * VCCattle_{t-1} * (1 + InflRate_t) + (BullsBuy_t * PriceCulledBulls_t) \tag{B.30}$$

This equation is an expansion on the formula used for livestock and was presented earlier as Eq. (10). Similar detailed equations are in place for each livestock type.

Total cash receipts for livestock on the farm are simulated by summing equations similar to Eq. (29) across all livestock types or:

$$\widetilde{L}R_t = \widetilde{C}RCattle_t + \widetilde{C}RSwine_t + \widetilde{C}RSheep_t + \widetilde{C}RGoats_t + \widetilde{C}RChickens_t \tag{B.31}$$

Total cash receipts from the sale of crops and livestock are simulated as:

$$\widetilde{T}Rec_t = \sum_i \widetilde{C}R_{it} + \widetilde{L}R_t \tag{B.32}$$

Total cash expenses for the farm family are the sum of variable costs for crops and livestock plus total fixed costs or:

$$\widetilde{T}Cost_t = \sum_i \widetilde{V}C_{it} + FC_t + \widetilde{V}CCattle_t + \widetilde{V}CSwine_t + \widetilde{V}CSheep_t + \widetilde{V}CGoats_t + \widetilde{V}CChickens_t \tag{B.33}$$

Net cash farm income for the farm family is a KOV for comparing the base and alternative technology scenarios and is simulated as:

$$NCFI_t = \widetilde{T}Rec_t - \widetilde{T}Cost_t \tag{B.34}$$

The cash flow for the family is a KOV for determining how an alternative technology scenario compares to the base scenario. Ending cash on December 31st is the objective for simulating the farm family's cash flow.

Cash inflows are simulated as:

Beginning Cash Jan 1 for t = 1 is given by the user; in subsequent years it is calculated as:

$$Beginning\ \ Cash_t = MAX(0,\ \ Ending\ \ Cash_{t-1})\ \ for\ \ years\ \ t = 2,\ \ 3,\ \ 4,\ \ and\ \ 5 \tag{B.35}$$

$$Interest\ \ Earned_t = Beginning\ \ Cash_{Jan1,t} * Interest\ \ Rate_t \tag{B.36}$$

$$\widetilde{C}ashInflow_t = BeginningCash_{Jan1,t} + \widetilde{N}CFI_t + \widetilde{I}nterestEarned_t \tag{B.37}$$

$$\widetilde{F}oodPurchased_t = \sum_i (PriceC_{it} * DeficitFoodPurchasedforCrop_{it}) + \sum_j (PriceL_{jt} * DeficitFoodPurchasedforLivestockType_{jt}) \tag{B.38}$$

Cash outflows include principle payments, taxes, and expenses for food purchased and other family living expenses or:

$$\widetilde{C}ashOutflow_t = PrinciplePayments_t + IncomeTaxes_t + SchoolExpenses_t + \widetilde{F}oodPurchased_t + FamilyLivingExpenses_t \tag{B.39}$$

If there is insufficient cash to cover all purchases of deficit food supplies, the model uses all available cash to purchase the possible amount of food.

Ending cash for the farm family is:

$$Ending\ \ Cash_t = Cash\ \ Inflow_t - Cash\ \ Outflow_t \tag{B.40}$$

FARMSIM simulates the variables to construct a balance sheet for the farm family. Assets include cash on hand, land, machinery, and livestock.

$$\widetilde{A}ssets_t = MAX(0, \widetilde{E}ndingCash_t) + LandValue_t + MachineryValue_t + \widetilde{L}ivestockValue_t \tag{B.41}$$

$$LandValue_t = LandValue_{t-1} * (1 + InflRate_t) \tag{B.42}$$

$$MachineryValue_t = MachineryValue_{t-1} * (1 + InflRate_t) \tag{B.43}$$

$$\widetilde{L}ivestockValue_t = \sum_j (\widetilde{L}Price_{jt} * No.\ Head_{jt}) \tag{B.44}$$

$$\widetilde{L}iabilities_t = LandDebt_t + MachineryDebt_t + LivestockDebt_t + MIN[0, \widetilde{E}ndingCash_t * -1.0)] \tag{B.45}$$

Debt and principal payments for the different assets are calculated in separate amortization tables to get annual interest and principal payments and remaining debt on December 31.

Net worth is calculated as:

$$\widetilde{N}etWorth_t = \widetilde{A}ssets_t - \widetilde{L}iabilities_t \tag{B.46}$$

Present value of ending net worth is used as a KOV:

$$\widetilde{P}VENW = NetWorth_5 * [1/(1 + DR)^5] \tag{B.47}$$

DR is the discount rate to convert future values to present year values.

$$ValueFarmProductsConsumed_t = \sum_i (PriceC_{it} * QuantityCropiConsumed_t) + \sum_j (PriceL_{jt} * QuantityLivestockjConsumed_t) \tag{B.48}$$

Net present value is simulated as:

$$\widetilde{N}PV = -BeginningNetWorth_{t=0} + [\sum_t (FamilyLivingExpenses_t + \widetilde{V}alueFarmProductsConsumed_t) * (1/(1 + DR)^t] - \widetilde{P}VENW \tag{B.49}$$

Nutrition calculations for the family extend FARMSIM beyond traditional farm budget and whole farm simulation models. The nutritional values for all crops and livestock products (meat, milk, and eggs) consumed by the family are simulated using FAO's nutrient values for each crop and livestock product (FAO, 2001a, 2001b, 2010), based on their average content of protein, calories, fat, iron, calcium and vitamin A. The formula to simulate available protein intake for a family is:

$$\begin{aligned}\widetilde{P}rotein_t = {} & [\sum_i (FamilyConsumption_{it} + PurchasedCrop_{it}) * GramsofProtein/UnitofCropi] \\ & + [\sum_j (FamilyConsumption_{jt} + PurchasedLivestockProduct_{jt}) * GramsofProtein/UnitofLivestockProduct_j]\end{aligned} \tag{B.50}$$

Eq. (B.50) is repeated for each of the remaining nutrient categories of: calories, fat, iron, calcium and vitamin A.

Probability of positive net cash farm income is calculated for each year t over the 500 iterations (s) as:

$$P(NCFI_t) = \sum_S (1 if NCFI_{ts} > 0, else 0)/500 \tag{B.51}$$

Probability of a cash flow deficit is a counter variable summed over the 500 stochastic iterations and is calculated for each year. A separate probability is calculated for each year over the 500 iterations (s) as:

$$P(CFD_t) = \sum_s (1 if EndingCash_{ts} < 0, else 0)/500 \tag{B.52}$$

Probability of economic success is the probability that NPV is greater than zero. If NPV is positive then the farm family's internal rate of return exceeds the discount rate (DR). The probability is the number of times NPV is positive over the 500 iterations (s) or:

$$P(ES_t) = \sum_s (1 if NPV_s > 0, else 0)/500 \tag{B.53}$$

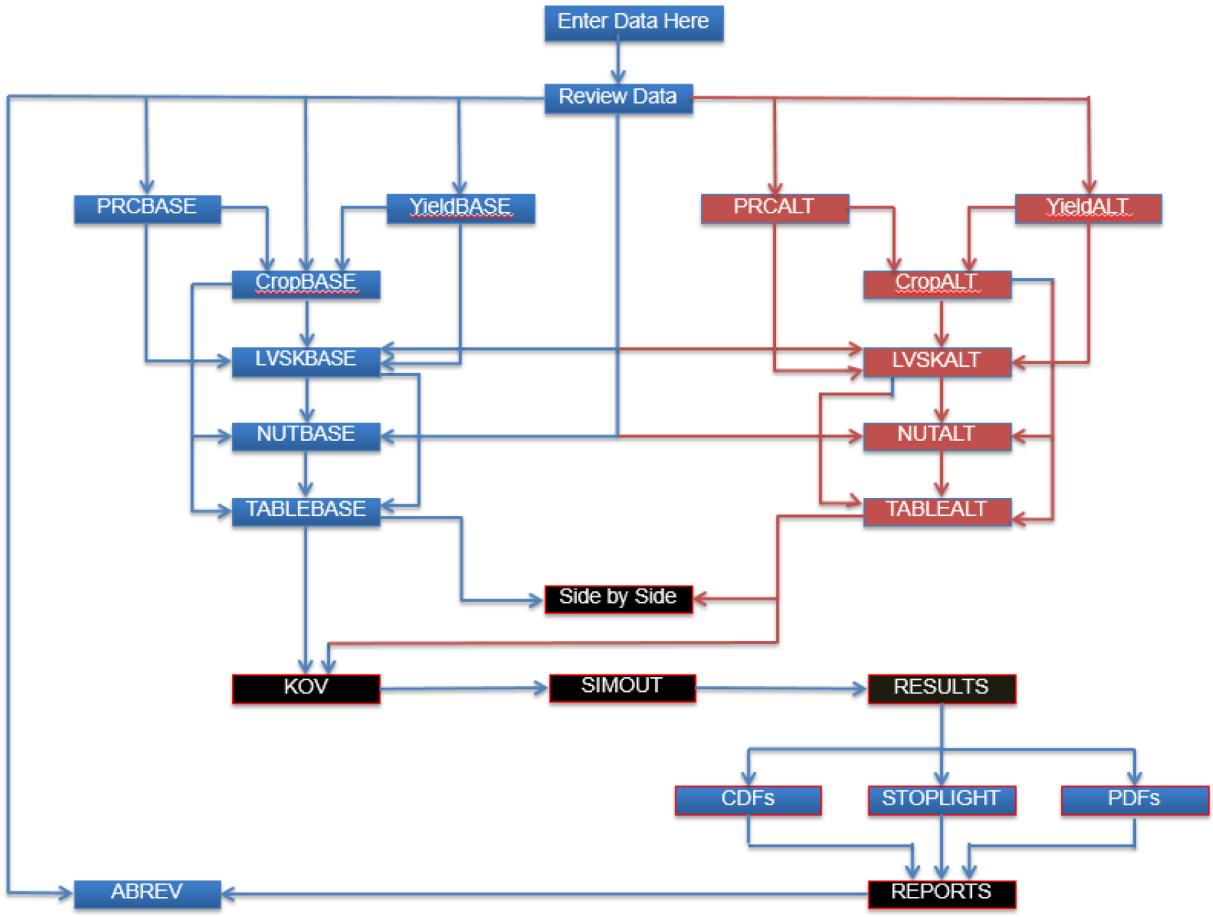
Probability that the farm family increased its real net worth is the number of times that PVENW exceeds beginning net worth over the 500 iterations (s) or:

$$P(IRNW) = \sum_s (1 if PVENW_s > BegNetWorth, else 0)/500 \tag{B.54}$$

Probability that the farm family's nutritional intake exceeds the FAO recommended daily requirements is calculated annually over the 500 iterations (s) for each of the six nutrient categories. The formula for each nutrient is the same as the equation for protein.

$$P(Protein_t) = \sum_s (1 if Protein_{ts}/365 > DailyMinimumReq, else 0)/500 \tag{B.55}$$

Appendix C:. FARMSIM flowchart (excel worksheet organization)



Abbreviations and meaning of different worksheets in FARMSIM:

YIELDBASE: stochastic yield worksheet in baseline scenario
PRCBASE: price worksheet in baseline scenario
CropBASE: crop worksheet in baseline scenario
LVSKBASE: livestock worksheet in baseline scenario
NUTBASE: nutrition worksheet in baseline scenario
TABLEBASE: table worksheet in baseline scenario
Note: same meaning of the above worksheets applies to the alternative scenarios (CropALT…)
KOV: key output variable
SIMOUT: simulation output

Appendix D

Appendix D.1. Water lifting technologies: description and assumptions

To evaluate the benefits and costs of alternative irrigation technologies (pulley vs. rope and washer, motor, and solar pumps) this analysis explicitly considers the costs of the different technologies and the associated cropland that can be irrigated without water stress. The assessment is based on costs (operating and capital) and capacity of the water lifting technology (pumping rate) to irrigate available land for a given crop's water needs. The following assumptions are considered for the analysis:

(1) Number of persons and hours per irrigation day: 1–2 persons for 1.5 h
(2) Number of irrigation days per season, assuming the farmer irrigates every other day for three and a half months (January-mid April): 65
(3) Total number of hours of irrigation per season: 1.5*65 = 97.5 h/season
(4) Pumping rates (liter/min) for the different water lifting technologies are:
- Pulley & tank: 15 L/min; Rope and washer operated by hand: 15 L/min; Motor pump: 170 L/min; Solar pump: 16 L/min

The pumping or discharge rates were obtained from field data gathered during the period of March to June 2015 by IWMI on behalf of the ILSSI project. Based on SWAT model (Soil and Water Assessment Tool) simulation results (see below footnote 13 for details), it was determined that

enough ground water was available to satisfy irrigation needs for all four alternative scenarios. The irrigator's equation is used to estimate the total amount of water that can be delivered by each water lifting technology per unit land (see Martin, 2011).[12]

Irrigator's equation: Q*T = d*A

Q: flow or pumping rate (liters/min)
T: time (min) for irrigation
d: depth of irrigation water applied (mm)
A: area covered (m^2 or ha)

Knowing the total amount of water (mm) required to irrigate a crop for the entire dry season and the total amount of water delivered by each water lifting technology per hectare (based on pumping rate and irrigation hours), we computed the fraction of water supply provided by each technology. Given the total irrigable land available for a crop (e.g. tomato) and its water requirements, we use the fraction of water supply by each technology to compute the fraction of available cropland that can be irrigated based on optimal water required to grow tomato for each water lifting technology.

Appendix D.2. Water lifting technology (WLT) characteristics, Robit kebele

Types of WLT	Operated by	Flow rate (l/min)	Cost WLT (ETB)	Additional Irrigable Land covered (ha)	Constraints
Pulley/tank	Hand	15	1310	411	labor
Rope & washer pump	Hand	15	3700	411	breakdowns
Motor pump	Fuel	170	8500	787	maintenance
Solar pump	Solar	16	16,000	438	capital costs

Note: we did not include the cost of digging wells in final estimation given that field data collected by IWMI in 2015 on behalf of ILSSI project shows that several households had a well.

Appendix E

Appendix E.1. Crop mix and land allocation (ha) scenarios for Robit kebele

Scenarios	Millet	Teff	Maize Wet season	Chickpeas	Potato	Irrigated[a] Cabbage Dry season	Irrigated Tomato	Irrigated Fodder	Irrigated Napier	Total (ha)
Baseline	708.0	266.0	728.0	57.0	24.0	126.0	102.0	43.0	43.0	2097.0
Alt.1 Pulley & tank	708.0	266.0	728.0	110.0	50.0	228.0	204.0	145.0	63.0	2502.0
Alt. 2 Rope & Washer Pump	708.0	266.0	728.0	110.0	50.0	231.0	207.0	148.0	63.0	2511.0
Alt. 3 Motor Pump	708.0	266.0	728.0	110.0	50.0	356.0	332.0	273.0	103.0	2926.0
Alt. 4 Solar Pump	708.0	266.0	728.0	110.0	50.0	240.0	216.0	157.0	63.0	2538.0

[a]Total potential irrigable land in Robit is 787 ha (source: SWAT[13] simulation results).

Appendix E.2. Current and recommended annual application rates of urea and DAP, Robit

	Fertilizer rates: Urea + DAP (kg/ha)		
Crops	Baseline scenario	Alternative scenario	
	current	Applied	Recommended
Teff	124	200	200
Maize	200	200	200
Millet	140	100	100
Tomato	56	300	350
Cabbage	8	140	250–300
Chickpeas	0	26	–
Potato	23	160	350
Fodder (oats & vetch)	0	100	–
Napier grass	0	100	–

[12] Martin (2011). Determining the Amount of Irrigation Water Applied to a Field. Arizona Water Series No. 29. Arizona Cooperative Extension, College of Agriculture and Life Sciences, University of Arizona.Available at: http://extension.arizona.edu/sites/extension.arizona.edu/files/pubs/az1157.pdf.

[13] SWAT (Soil and Water Assessment Tool) model results (p.9) at: http://ilssi.tamu.edu/media/1333/final-bdz-robit-ex-ante-analysis-of-ssi-scenarios.pdf.

Appendix E.3. Mean crop yields (kg/ha), land are (ha) and input costs (Birr/ha) for scenarios in Robit

	Baseline scenario						Alternative scenarios					
Crops	Mean yield (kg/ha)	Crop area /hh (ha)	Cost fert. (Birr/ha)	Cost seed (Birr/ha)	Cost irrig labor (Birr/ha)	Other labor cost (Birr/ha)	Mean yield (kg/ha)	Crop area /hh (ha)	Cost fert. (Birr/ha)	Cost seed (Birr/ha)	Cost irrig labor (Birr/ha)	Other labor cost (Birr/ha)
Teff	838	0.13	1614	470	0	172	1995	0.13	4800	470	0	172
Maize	2127	0.36	4284	476	258	73	2773	0.36	4284	476	258	1350
Millet	1640	0.35	3110	46	0	0	2257	0.35	3110	46	0	1350
Tomato	14,293	0.05	783	420	259	250	21,714	0.10	1875	420	10,757	4181
Cabbage	11,376	0.06	110	880	258	200	18,089	0.11	110	880	10,757	600
Chickpeas	1274	0.02	358	122	258	0	1274	0.05	358	122	258	0
Potato	3770	0.01	100	0	736	0	7728	0.02	1504	2595	736	1504
Fodder (O & V)	12,654	0.02	0	300	258	0	34,168	0.07	3000	1200	10,757	0
Napier grass	10,936	0.02	926	234	259	1482	10,936	0.03	926	234	10,757	1482

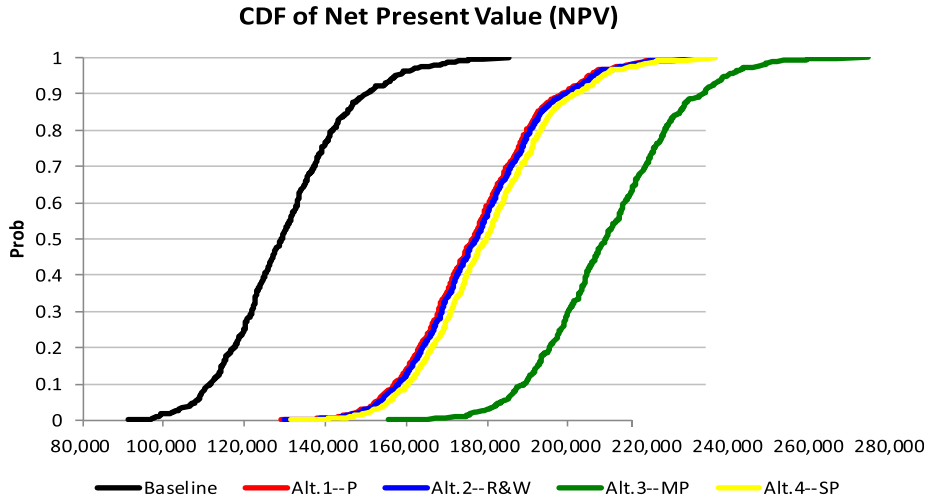
Notes: (1) fert. = fertilizer; irrig. = irrigation; hh = household; O & V = Oats & Vetch.
(2) Increase in "Other labor cost" in alternative scenarios are due to increased labor costs of breaking down the hardpan soil

Appendix F. . Input variables and livestock technology scenarios in Robit kebele

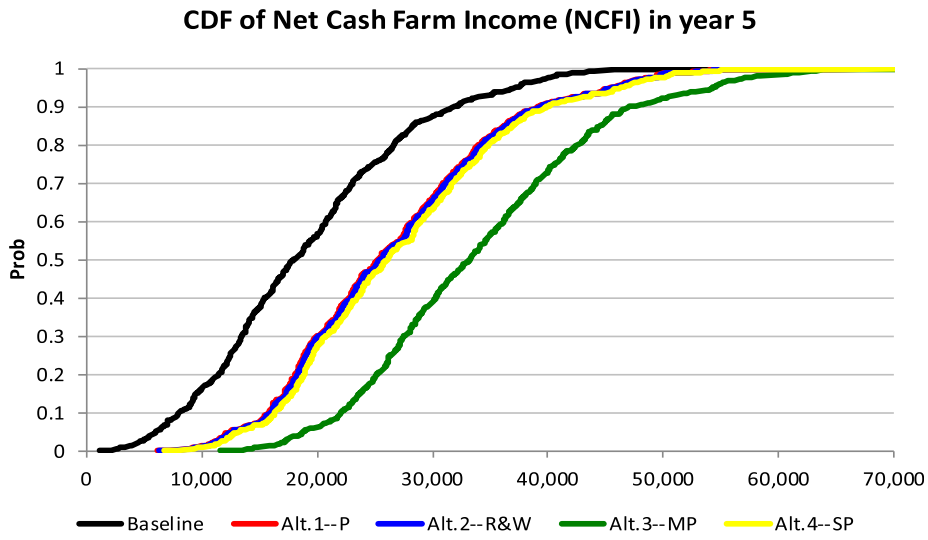
	Baseline scen.	Alt. scen. 1	Alt. scen. 2	Alt. scen. 3	Alt. scen. 4
Cows					
Native	2640	2640	2640	2640	2640
Cross-breeds	165	165	165	165	165
Milk per cow					
Liters/cow/year	185	312	312	312	312
Live Weight gain (kg)	0	52.4	52.4	52.4	52.4
Live weight /bull	184	236.4	236.4	236.4	236.4
Consumption			Percent (%)		
Milk by family	28	38	38	38	38
Milk by employees	0	0	0	0	0
Made into butter	70	50	50	50	50
Butter sold	54	54	54	54	54

Appendix G

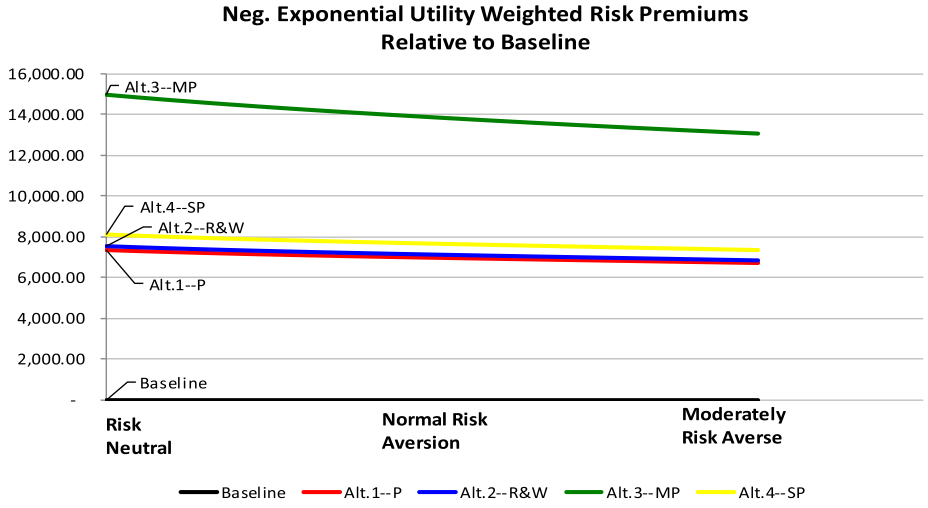
Appendix G.1. CDF of NPV for alternative irrigation technologies in Robit kebele



Appendix G.2. CDF of NCFI for base and alternative irrigation technologies in Robit kebele



Appendix H. . Risk premiums ranking of alternative farming systems in Robit kebele



Appendix P. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2018.11.038.

References

Agueh, V.D., Tugoué, M.F., Sossa, C., Métonnou, C., Azandjemè, C., Paraiso, N.M., Ouendo, M.-E., Ouédraogo, L.T., Makoutodé, M., 2015. Dietary Calcium intake and Associated Factors among Pregnant Women in Southern Benin in 2014. Food Nutr. Sci. 6, 945–954. https://doi.org/10.4236/fns.2015.611098.

Ahmed, M.H., 2015. Adoption of multiple agricultural technologies in maize production of the Central Rift Valley of Ethiopia. Stud. Agric. Econ. 117, 162–168. https://doi.org/10.7896/j.1521.

Arimond, M., Ruel, M.T., 2004. Community and International Nutrition Dietary Diversity Is Associated with Child Nutritional Status: evidence from 11 Demographic and Health Surveys 1, 2. J. Nutr. 134, 2579–2585.

Barrett, C.B., 2005. Spatial Market Integration. The New Palgrave Dictionary of Economics. In: Blume, Lawrence E., Durlauf, Steven N. (Eds.), Second ed. Palgrave Macmillan, London, Forthcoming. Available at SSRN: < https://ssrn.com/abstract=1142520 > .

Bizimana, J.C., Angerer, J.P., Bessler, D.A., Keita, F., 2015. Cattle markets integration and price discovery: the case of Mali. J. Develop. Stud. 51 (3), 319–334. https://doi.org/10.1080/00220388.2014.963564.

De Janvry, A., Dustan, A., Sadoulet, E., 2011. Recent Advances in Impact Analysis Methods for Ex-post Impact Assessments of Agricultural Technology: Options for the CGIAR. SPIA report version 3.1. University of California at Berkeley.

Di Falco, S., Chavas, J., 2009. On crop biodiversity, risk exposure, and food security in the highlands of Ethiopia. Am. J. Agric. Econ. 91 (3), 599–611.

Diagne, A., Glover, S., Groom, B., Phillips, J., 2012. Africa's Green Revolution? The determinants of the adoption of NERICAs in West Africa. SOAS Department of Economics Working Paper Series, No. 174, SOAS, University of London.

Diao, X., Hazell, P., Thurlow, J., 2010. The role of agriculture in African Development. World Dev. 38 (10), 1375–1383. https://doi.org/10.1016/j.worlddev.2009.06.011.

Dile, Y.T., Srinivasan, R., 2014. Evaluation of CFSR climate data for hydrologic prediction in data-scarce watersheds: an application in the Blue Nile River Basin. J. Am. Water Resour. Assoc. (JAWRA), pp. 1–16. 10.1111/jawr.12182.

Dorosh, P., Thurlow, J., 2013. Agriculture and small towns in Africa. Agric. Econ. 44 (4–5), 449–459. https://doi.org/10.1111/agec.12027.

Doss, C.R., 2006. Analyzing technology adoption using microstudies: limitations,

challenges, and opportunities for improvement. Agric. Econ. 34, 207–219. https://doi.org/10.1111/j.1574-0864.2006.00119.x.
FAO (Food and Agriculture Organization of the United Nations), 2001a. Human Energy Requirements. FAO/WHO/UNU Expert Consultation. Food and Nutrition Technical Report Series No.1. Food and Agricultural Organization of the United Nations, Rome, Italy.
FAO (Food and Agriculture Organization of the United Nations, 2001b). Human Vitamin and Mineral Requirements. Report of a Joint FAO/WHO Expert Consultation. Nutrition Division, Food and Agricultural Organization of the United Nations, Rome, Italy.
FAO (Food and Agriculture Organization of the United Nations), 2010. Fats and Fatty Acids in Human Nutrition. Report of an Expert Consultation. Food and Agricultural Organization of the United Nations, Rome, Italy.
FAO (Food and Agriculture Organization of the United Nations), 2011. Minimum Dietary Energy Requirements. Available online and accessed on 2018-06-18 at: < http://www.fao.org/fileadmin/templates/ess/documents/food_security_statistics/MinimumDetaryEnergyRequirement_en.xls > .
Federal Democratic Republic of Ethiopia, 2008. Summary and Statistical Report of the 2007 Population and Housing Census. Published by the Population Census Commision, Addis-Ababa, Ethiopia. Available at: < https://www.scribd.com/doc/28289334/Summary-and-Statistical-Report-of-the-2007 > .
FEWS NET (Famine Early Warning System Network), 2015. Illustrating the extent and severity of the 2015 drought. Ethiopia Special Report, December 17, 2015. Available at: < http://fews.net/sites/default/files/documents/reports/FEWS%20NET_WFP_Ethiopia%20Alert_20151204.pdf > .
Gebregziabher, G., Rebelo, L-M., Notenbaert, A., Ergano, K., Abebe, Y., 2013. Determinants of Adoption of Rainwater Management Technologies among Farm Households in the Nile River Basin. IWMI Research Report 154. Available at: < http://www.iwmi.cgiar.org/Publications/IWMI_Research_Reports/PDF/pub154/rr154.pdf?galog=no > .
Gebremedhin, B., Hoekstra, D., Tegegne, A., Shiferaw, K., Bogale, A., 2015. Factors determining household market participation in small ruminant production in the Highlands of Ethiopia.
Gebremeskel, D.K., 2016. Assessment of production practices and effect of N:P2O5:S rates on yield and yield components of head cabbage (brassica oleracea var. capitata) under irrigation conditions in Lay Armachio district, Amhara region, Ethiopa. M.Sc. Thesis, Bahir Dar University, Unpublished. Available at: < https://cgspace.cgiar.org/bitstream/handle/10568/90441/Kidane.pdf?sequence=1 > .
Hardaker, J.B., Richardson, J.W., Lien, G., Schumann, K.D., 2004a. Stochastic efficiency analysis with risk aversion bounds: a simplified approach. Aust. J. Agric. Resour. Econ. 48 (2), 253–270.
Hardaker, J., Huirne, R., Anderson, J.R., Lien, G., 2004b. Coping with Risk in Agriculture. CABI Publishing, Oxfordshire, OX.
Key, N., Sadoulet, E., de Janvry, A., 2000. Transactions costs and agricultural household supply response. Am. J. Agric. Econ. 82 (2), 245–259. https://doi.org/10.1111/0002-9092.00022.
Norton, G., 2014. Hunger and Hope: Escaping Poverty and Achieving Food Security in Developing Countries. Waveland Press Inc., Long Grove, IL.
Qasim, M., 2012. Determinants of Farm Income and Agricultural Risk Management Strategies: The Case of Rain-Fed Farm Households in Pakistan's Punjab. Kassel University Press GmbH.
Richardson, J.W., Herbst, B.K., Outlaw, J.L., Gill, R.C., 2007. Including risk in economic feasibility analyses: the case of ethanol production in texas. J. Agribus. 25 (2), 115–132.
Richardson, J.W., Klose, S.L., Gray, A.W., 2000. An applied procedure for estimating and simulating multivariate empirical (MVE) probability distributions in farm-level risk assessment and policy analysis. J. Agric. Appl. Econ. 32 (2), 299–315.
Richardson, J.W., Nixon, C.J., 1985. FLIPSIM V: A General Firm Level Policy Simulation Model. B-1528. Department of Agricultural Economics, Texas A & M University, USA.
Shetty, P., 2010. Addressing micronutrient malnutrition to achieve nutrition security. In: Thompson, B., Amoroso, L. (Eds.), Combating micronutrient deficiencies: food-based approaches. FAO and CAB. 10.1079/9781845937140.0000.
USAID (United States Agency for International Development), 2011. Ethiopia: Feed The Future Multi-Year Strategy. Accessed in October 2015 and available at: https://www.usaid.gov/sites/default/files/documents/1860/USAID%20FtF%20MYS%20Final%20Version.pdf.
USAID (United States Agency for International Development), 2013. Cost-benefit analysis of the potatoes, onions, and tomatoes value chains in Ethiopia. Graduation with resilience to achieve sustainable development (grad) project. Final report.
von Grebmer, Klaus, Bernstein, Jill, Nabarro, David, Prasai, Nilam, Amin, Shazia, Yohannes, Yisehac, Sonntag, Andrea, Patterson, Fraser, Towey, Olive, Thompson, J., 2016. 2016 Global Hunger Index: The Getting to zero hunger.
Williams, J.R., Arnold, J.G., Srinivasan, R., Ramanarayanan, T.S., 1998. APEX: A new tool for predicting the effects of climate and CO_2 changes on erosion and water quality. In: Boardman, J., Favis-Mortlock, D. (Eds.), Modelling Soil Erosion by Water. Springer-Verlag, Heidelberg, pp. 441–449.
Wondatir, Z., Adie, A., Duncan, A.J., 2015. Assessment of livestock production and feed resources at Robit Bata, Bahir Dar, Ethiopia. ILSSI report: Nairobi, Kenya, ILRI. Available at: < https://cgspace.cgiar.org/handle/10568/56988 > .
Yengoh, G.T., Armah, F.A., Svensson, M.G., 2009. Technology adoption in small-scale agriculture. Sci., Technol. Innov. Stud. 5 (2), 111–131.
Zemadim, B., McCartney, M., Sharma, B., Wale, A., 2011. Integrated rainwater management strategies in the blue nile basin of the ethiopian highlands. Int. J. Water Resour. Environ. Eng. 3 (10), 220–232.